Feed-om of Speech: The Technical Design and Legal Interpretation of Social Media Recommender Systems

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Table of Contents

Chapter 1: Introduction 6
An Evolving Information Ecosystem 6
Developing Preferences and Recommendations 7
Incentives for Engagement 8
Legal and Regulatory Efforts 11
Chapter 2: Related Works 13
Technical Studies of Recommender Systems 13
Legal Classification of Recommender Systems 15
Chapter 3: The Technical Representation of Content 18
Signals 18
Features 22
Chapter 4: The Recommendation Itself 28
Recommender System Architecture 28
Collaborative Filtering 30
Content-Based Filtering 33
Advanced Architecture 35
Two-Tower Systems 35
Multimodal systems 36
Deep Learning 37
Model Parameters 39
Hyperparameter Tuning 40
Ranking Algorithms 43
Chapter 5: The Legal Classification of Recommender Systems 47
Origins and Intent of Section 230 48
Early Interpretations of Section 230 50
Algorithmic Curation under Section 230 54
Editorial Control and First Amendment Protection 56
Evolving Interpretations 63
Chapter 6: Remedies 70

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Any remaining errors are, of course, my own.

Dedication

To my grandparents, Renee, Judy, Herb, and Jerry, whose memories will forever be a blessing, and who inspire me always to look at the world with a high level of dimensionality.

Chapter 1: Introduction

An Evolving Information Ecosystem

The social media feed is a ubiquitous part of American life today.[1] Facebook, Instagram, TikTok or X; in the morning, midday or evening; on your phone, tablet or computer. Scrolling through whatever posts have popped up on your personalized feed has become a cultural ritual. There are many folk theories about the algorithms behind the feed, but these theories generally pertain more to output (feed) rather than implementation (system).[2] Growing concerns around potential harms of social media and the proprietary nature of the underlying recommender systems have created an information ecosystem saturated with confusion and anxiety. Most users lack control over what they see, and technical opacity precludes them from even understanding why they see what they see. Policymakers, researchers, and the public are left speculating about the root causes of polarization, misinformation, and other harms.

At its origins, social media was an active venture in that it required users to look up their friends or communities to participate in forums or chatrooms. Development of the feed introduced passivity to social media—a user could sit back and simply scroll—no search necessary. Early on, users had near-complete agency over their feeds. The Facebook News Feed, which debuted in 2006, showed posts and activity by Facebook friends of the user. If a user did not want to see another user on their feed, they could simply unfriend them. Gradually, Facebook implemented feedback features to further increase user control such as indicating which accounts they did and did not want to see posts from.

However, by 2011 the concept of user agency had shifted. Facebook began displaying content based on popularity and individual user engagement prediction rather than simply chronologically or based on Friends lists. Today, Facebook's over three billion daily users see unique feeds based on their own individual preferences, including posts by accounts and organizations the user has never interacted with.[3] "User preference" has gone a thousand clicks

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[1] Jeffrey Gottfried, *Americans' Social Media Use*, Pew Rsch. Ctr. (Jan. 31, 2024), https://www.pewresearch.org/internet/2024/01/31/americans-social-media-use/.
[2] Brita Ytre-Arne & Hallvard Moe, *Folk Theories of Algorithms: Understanding Digital Irritation*, 43 Media, Culture & Soc'y, no. 5 (Dec. 2020), https://doi.org/10.1177/0163443720972314.
[3] Stacy J. Dixon, *Biggest Social Media Platforms by Users 2024*, Statista (2024), https://www.statista.com/statistics/272014/global-social-networks-ranked-by-number-of-users/.

beyond the original idea of showing users their friends. As platforms discovered the large profit potential from advertising revenue, they built complex technical recommender systems to automate algorithmic curation across vast user bases and optimize for engagement and profit. The systems are technical, but they are built on human assumptions about behavior and preference.

Developing Preferences and Recommendations

When we eat at a restaurant, we evaluate the experience using our senses—taste buds detect flavor, eyes take in ambiance through rods and cones, and eardrums process sound waves from the server's tone. We decide whether to recommend the restaurant to others based on these sensory inputs, filtered through personal experience, social influence, and emotional response. Meanwhile, computers can also "see" and "hear" using technologies like computer vision and speech recognition, and they, like humans, rely on inputs to evaluate experiences. The core distinction between humans and computers lies in how those inputs are processed. Humans continuously and unconsciously form preferences based on stimuli through personal experience, instinct, emotional context, and cultural cues.[4] These mechanisms are messy, inconsistent, and deeply organic.

By contrast, artificial intelligence systems do not form preferences organically, rather, their preferences are shaped by engineering choices around software architecture, training data, and other technical processes. A well-known artificial intelligence (AI) textbook defines artificial intelligence as "the study of agents that receive percepts from the environment and perform actions."[5] The same textbook categorizes two ways to measure the success of AI: in terms of fidelity to human performance, and against an ideal performance measure (rationality).[6] Both measures are qualitative, and what one person views as successful or rational is subjective. When AI systems learn information to perform tasks, they rely on their engineers' definition of success to guide their behavior.

In our restaurant analogy, if a computer is trained to prioritize relevant factors like taste, sound, and service quality, it's likely that the computer and the human may reach similar conclusions about the restaurant. However, human preference is messy and innate, while a computer's preferences are determined by the objectives and constraints defined by its

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[4] Michel Tuan Pham, *Emotion and Rationality: A Critical Review and Interpretation of Empirical Evidence*, 11 Review of General Psychology 2 (2007), https://doi.org/10.1037/1089-2680.11.2.15.
[5] Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach* viii (3d ed. 2010).
[6] *Id.* at 1.

programmers. A computer can only "care" about what it has been trained to care about. Human preference develops more organically over time.

Another human source of influence is peer-based. Imagine that instead of relying on personal experience, you evaluate the restaurant based solely on how other diners seemed to be reacting. If you see others enjoying their food, you will assume the food was good. If you thought others were unsatisfied, you would decide the food was bad—even if you didn't dislike it yourself.[7] Peer-based evaluation is a method both humans and computers are capable of. Just as a person's opinion may be based on social cues as proxies for direct experience, a social media recommender system collects cues as well. It generally relies on engagement data (likes, shares, time spent watching a piece of content) as proxies for quality or relevance. For both humans and computers, relying on external cues can flatten the complexity of an experience into simplified impressions. However, humans have the capacity to reflect, adapt, and reconsider in ways that most computer systems must be prompted to do.

This is how social media platforms make recommendations today. Recommender systems process complex user engagement data for algorithmic curation by flattening the focus to how users are reacting to content rather than the quality of the content itself. Systems behave this way because platforms rely on advertising revenue, which depends on engagement. With their exponential proliferation and social dependence, recommender systems have come to play a central role in shaping user experiences. Their increasing complexity and influence raise complex legal questions that may only be answered reasonably with an understanding of how the systems work.

Incentives for Engagement

The primary objective of social media platforms' recommender systems is "to rank the available content according to how likely it is that the user in question will engage with it."[8] This

[7] On short-form social media, where content is viewed for a few seconds to a few minutes, a user's initial reaction is their only reaction. In the restaurant analogy, this would be like forming an opinion about a restaurant based only on a five-second scan of other diners' reactions.

[8] Arvind Narayanan, *Understanding Social Media Recommendation Algorithms*, Knight First Amend. Inst. at Columbia Univ. (2023), https://knightcolumbia.org/content/understanding-social-media-recommendation-algorithms.

aligns with the business structure of the platforms, which benefits when users remain engaged for longer periods of time because of longer exposure to targeted advertising.

Financial documents and an examination of advertising algorithms demonstrate the relationship between engagement and revenue. Fiscal reporting exposes the level to which social media companies are reliant on advertising. In its 2024 Q1 10-Q SEC filing, Alphabet notes that it generates "a significant portion of our revenues from advertising. Reduced spending by advertisers, a loss of partners, or new and existing technologies that block ads online and/or affect our ability to customize ads could harm our business."[9] Advertising generated over seventy-five percent of Alphabet's total revenues in 2023. From Meta's 2024 Q3 10-Q: "Substantially all of our revenue is currently generated from advertising on Facebook and Instagram."[10] Ninety-eight percent of Meta's revenue for the quarter came from advertising. The companies claim that because they rely on advertising revenue, they are obligated to deliver effective targeted ads to users to maintain their business models.

Engagement is important for generating advertising revenue because advertisers are willing to pay more for ad slots that are likely to generate high engagement, such as clicks or conversions. Internet advertising has two key components: who sees the ad and how many people see it. To determine the first part, advertising platforms have hundreds of thousands of ways to label users. Labels may be based on information the user has no idea they've given up, for example, whether they are tattoo-addicted, an overindulgent dog owner, or live in a household with a trendy mom.[11] These labels are based on users' engagement activity across the Internet and are aggregated and sold in an ever-growing marketplace of user data.[12] Engagement history matters because it indicates what users might buy in the future. Advertisers want to maximize overall sales, which brings us to the second part of Internet advertising: scale. Even perfectly targeted advertising profit would remain limited if it was only able to reach one individual. One person can only buy so much, so advertisers must personalize content *and* broaden their audience for maximum revenue.

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[9] Alphabet Inc., *Form 10-Q*, at 15 (Mar. 31, 2024), https://abc.xyz/assets/9c/12/c198d05b4f7aba1e9487ba1c8b79/goog-10-q-q1-2024.pdf.

[10] Meta Platforms, Inc., *Form 10-Q* (Sept. 30, 2024), https://www.sec.gov/Archives/edgar/data/1326801/000132680124000081/meta-20240930.htm.

[11] Jon Keegan & Joel Eastwood, *From "Heavy Purchasers" of Pregnancy Tests to the Depression-Prone: We Found 650,000 Ways Advertisers Label You*, The Markup (June 8, 2023), https://themarkup.org/privacy/2023/06/08/from-heavy-purchasers-of-pregnancy-tests-to-the-depression-prone-we-found-650000-ways-advertisers-label-you.

[12] Tim Mucci, *What Is a Data Marketplace?*, IBM (2024), https://www.ibm.com/think/topics/data-marketplace.

A key element of the online advertising landscape are auction-based mechanisms such as the Generalized Second-Price auction. These mechanisms determine both the placement and cost of advertisements. In these auctions, advertisers bid for ad slots. The most desirable slots are positioned in high-traffic areas, as they generate greater engagement and higher earnings.[13]

Key advertising inventory pricing models include Cost Per Click (CPC), Cost Per Thousand Impressions (CPM), and dynamic pricing mechanisms.[14] CPC and CPM explicitly use engagement to measure user interactions with advertisements. CPC only charges advertisers when users actively engage with an ad by clicking, while CPM charges advertisers based on the number of times an ad is displayed.[15] Both models rely on user engagement data to assess the effectiveness and value of ad placements. If there are more users active on the platform, there is more potential for ad placements which generate high CPC and CPM.

The cost of ad slots using CPC and CPM as value indicators is determined through dynamic pricing mechanisms such as real-time bidding (RTB) or other auction-based mechanisms. RTB relies on second-price mechanisms, where the highest bidder for a slot pays the price of the second-highest bid.[16] Also known as Vickrey auctions, RTB encourages truthful bidding and maximizes platform revenue because it ensures that valuable ad placements go to the entities willing to pay the most.

Dynamic pricing allocates ad slots based on real-time demand and market competition, both informed by engagement data. Prices for CPC can increase during peak hours or for highly competitive audiences. Prime slots, such as the top of the feed, are more visible and generate higher clicks, while less prominent slots receive lower visibility and interaction. Platforms use split-second RTB auctions to sell their slots efficiently, allowing advertisers to compete for the best placements.

As shown above, there is a clear incentive for platforms to maximize user engagement because engagement dictates the profitability of ad slots. According to former Google employee Guillaume Chaslot, "[The YouTube recommender system] isn't built to help you get what you

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[13] AgencyAnalytics, *Ad Revenue*, KPI Examples, https://agencyanalytics.com/kpi-definitions/ad-revenue (last visited Mar. 28, 2025).
[14] Nikita Bykadarov, *Understanding Ad Inventory: A Guide to Effective Management and Optimization*, Improvado (2025), https://improvado.io/blog/understanding-ad-inventory.
[15] Criteo, *What's the Difference Between CPC and CPM?* (2017), https://www.criteo.com/blog/whats-difference-cpc-cpm/.
[16] *Introduction to Real-Time Bidding (RTB) - Authorized Buyers Help*, Google Help, https://support.google.com/authorizedbuyers/answer/6136272?hl=en (last visited Feb. 24, 2025).

want—it's built to get you addicted to YouTube."[17] In other words, the recommender systems are designed to maximize engagement by any means necessary.

Legal and Regulatory Efforts

The rise of algorithmic recommender systems raises legal questions about whether algorithmic curation constitutes expressive, first-party speech protected by the First Amendment or simply neutral functional infrastructure covered by 47 U.S.C. § 230(c)(1). Enacted in 1996, Section 230 shields "interactive computer services" from liability for user-generated content. Section 230 was born when computer services largely did not employ personalized recommender systems. Instead, these services functioned as the technical infrastructure on which to house user generated content—for instance, America Online and CompuServe. When applied to social media, Section 230 offers platforms broad protections from liability for users' posts, regardless of unlawfulness or harm caused. On the other hand, treating platforms as newspaper editors and publishers affords them First Amendment protection from government intervention but leaves them liable for what they publish, if, for example, plaintiffs can prove the publisher knowingly published unlawful content.[18]

The opacity of social media platforms regarding their technical processes has hindered a complete technical understanding of recommender systems by external parties, including the government and the general public. It has obscured the degree of subjectivity with which such systems are created and maintained despite widespread hypotheses about their effects on polarization and misinformation.[19,20,21,22] With profit as the ultimate maximization target and user agency kept at a minimum, platforms have no reason to change the overarching behavior of the feed beyond its already enormously successful implementation.

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[17] Már Másson Maack, *'YouTube Recommendations Are Toxic,' Says Dev Who Worked on the Algorithm*, The Next Web (June 14, 2019), https://thenextweb.com/news/youtube-recommendations-toxic-algorithm-google-ai.
[18] *Miami Herald Publ'g Co. v. Tornillo*, 418 U.S. 241 (1974).
[19] Ari Shapiro, Michael Levitt & Christopher Intagliata, *How the Polarizing Effect of Social Media Is Speeding Up*, NPR (Sept. 9, 2022), https://www.npr.org/2022/09/09/1121295499/facebook-twitter-youtube-instagram-tiktok-social-media.
[20] Kofi Annan Found., *Navigating the Age of Polarization: How Social Media Fuels Divides* (2023), https://www.kofiannanfoundation.org/news/navigating-the-age-of-polarization-how-social-media-fuels-divides/.
[21] Sounman Hong & Sun H. Kim, *Political Polarization on Twitter: Implications for the Use of Social Media in Digital Governments*, 33 Gov't Info. Q. 777 (2016), https://doi.org/10.1016/j.giq.2016.04.007.
[22] Christian Montag & Simon Hegelich, *Understanding Detrimental Aspects of Social Media Use: Will the Real Culprits Please Stand Up?*, 5 Frontiers in Soc. 599270 (Dec. 2020), https://doi.org/10.3389/fsoc.2020.599270.

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Social media is the new American town square, a space where public discourse evolves in real time and personal expression is algorithmically curated. With a wider understanding of the algorithms and architecture that make up recommender systems, the regulatory and cultural landscape will be better equipped to move forward as an information-centric society. This thesis examines the technical characteristics of engagement-driven recommender systems by analyzing their algorithmic design. It then considers the legal consequences of classifying them as first-party speech—or, alternatively, as conduct that falls within the scope of Section 230 protection. If such systems are deemed first-party speech, recent jurisprudence suggests that Section 230 protections are irrelevant to the question of liability. If they are not, as precedent and much of the legal community believes, legal and policy discussions must turn to interpreting or amending Section 230—a question beyond the scope of this thesis. Either way, moving forward requires a strong bridge across legal and technical disciplines and a willingness to not see these systems as binary, but rather as existing along a spectrum of business development, editorial involvement, and algorithmic design.

Chapter 2: Related Works

The study of recommender systems extends across technical and legal perspectives. Technical researchers have developed increasingly sophisticated models to emulate platforms' data processing, feature selection, and ranking algorithms. Public research by the platforms themselves also offers some insight into the inner workings of their recommender systems. On the legal side, scholars and policymakers have focused on the implications of algorithmic curation for free speech and platform liability, particularly with regards to Section 230. Yet despite this rich body of research, there is relatively little scholarship that integrates these technical and legal analyses: This is the interdisciplinary gap this thesis aims to address.

Technical Studies of Recommender Systems

We find two broad types of research relating to recommender systems for social media. There is research focused on a particular low-level component of the systems, such as feature selection, and research which reviews recommender systems overall without going into depth about the granular technical processes. There are some exceptions to this categorization.

Additionally, there is an abundance of literature offering highly detailed, technical insights into various components of machine learning and recommender systems, such as those published in *Journal of Machine Learning Research* or presented at RecSys, that are not specific to social media. Much of this work focuses on isolated parts of the recommendation process, and as a result, may overlook how these components fit into larger system architecture or how they shape the overall behavior and outputs of the system.

There is some work that focuses on specific technical components, for instance, Jiliang Tang and Huan Liu have written extensively about feature selection within the context of social

media.[23,24,25,26,27] Recent work by Pengyue et al. highlights the importance of benchmarking feature selection and reproducing its effects on recommender systems.[28] These efforts contribute to a growing recognition that personalized recommender systems depend as much on data processing as they do on downstream modeling, and that approach to feature selection greatly impacts performance regardless of model choice.

System architecture, particularly ranking algorithms, are central in shaping user experience on social media. Several scholars have provided high-level overviews of recommender systems as a field. Burke et al.[29], Roy and Dutta[30], and Anandhan et al.[31] offer explanations of collaborative and content-based filtering systems. In their seminal text, Jannach et al. provide a thorough, top-down guide to recommender systems including evaluation techniques and recent developments.[32]

More recently, Paul Bouchaud has examined the economic and behavioral consequences of social media recommender systems, particularly how subtle changes in ranking logic can impact engagement metrics and online polarization.[33,34,35] Arvind Narayanan produced a number of useful

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[23] Jiliang Tang & Huan Liu, *Unsupervised Feature Selection for Linked Social Media Data*, in *Proceedings of the 18th ACM SIGKDD Int'l Conf. on Knowledge Discovery & Data Mining* 904 (Aug. 2012), https://doi.org/10.1145/2339530.2339673.

[24] Jiliang Tang, Huiji Gao & Huan Liu, *Unsupervised Feature Selection for Multi-View Data in Social Media*, in *Proceedings of the 2013 SIAM Int'l Conf. on Data Mining* 270 (2013), https://doi.org/10.1137/1.9781611972832.30.

[25] Jiliang Tang, Salem Alelyani & Huan Liu, *Feature Selection for Classification: A Review*, in *Data Classification* 37 (CRC Press 2014), https://doi.org/10.1201/b17320.

[26] Jiliang Tang & Huan Liu, *Feature Selection for Social Media Data*, 8 ACM Transactions on Knowledge Discovery from Data, no. 4, art. 19 (Oct. 2014), https://doi.org/10.1145/2629587.

[27] Jungdong Li et al., *Unsupervised Streaming Feature Selection in Social Media*, in *Proceedings of the 24th ACM Int'l Conf. on Information & Knowledge Mgmt.* 1041 (Oct. 2015), https://doi.org/10.1145/2806416.2806501.

[28] Pengyue Jia et al., *ERASE: Benchmarking Feature Selection Methods for Deep Recommender Systems*, in *Proceedings of the 30th ACM SIGKDD Conf. on Knowledge Discovery & Data Mining* 5194 (Aug. 2024), https://doi.org/10.1145/3637528.3671571.

[29] Robin Burke, Alexander Felfernig & Mehmet H. Göker, *Recommender Systems: An Overview*, 32 Ass'n for the Advancement of Artificial Intelligence, no. 3, at 13 (2011).

[30] Deepjyoti Roy & Mala Dutta, *A Systematic Review and Research Perspective on Recommender Systems*, 9 J. Big Data 59 (2022), https://doi.org/10.1186/s40537-022-00592-5.

[31] Anitha Anandhan et al., *Social Media Recommender Systems: Review and Open Research Issues*, 6 IEEE Access (2018), https://doi.org/10.1109/ACCESS.2018.2810062.

[32] Dietmar Jannach, Markus Zanker & Gerhard Friedrich, *Recommender Systems: An Introduction* (Cambridge Univ. Press 2011).

[33] Paul Bouchaud, David Chavalarias & Maziyar Panahi, *Crowdsourced Audit of Twitter's Recommender Systems*, 13 Sci. Reps. 16815 (2023), https://doi.org/10.1038/s41598-023-43980-4.

[34] Paul Bouchaud, *Algorithmic Amplification of Politics and Engagement Maximization on Social Media*, in *Complex Networks & Their Applications XII* 145 (Hocine Cherifi et al. eds., vol. 1144, Springer 2024), https://doi.org/10.1007/978-3-031-53503-1_11.

[35] Paul Bouchaud, *Skewed Perspectives: Examining the Influence of Engagement Maximization on Content Diversity in Social Media Feeds*, 7 J. Computational Soc. Sci. 721 (Mar. 2024), https://doi.org/10.1007/s42001-024-00255-w.

reports on the overall design of social media platforms.[36,37,38] His work and complementary analyses from the Knight First Amendment Institute's Algorithmic Amplification and Society project have explored how algorithms exploit and dominate our attention.[39,40]

Legal Classification of Recommender Systems

Legal scholars have increasingly questioned how algorithmic recommendation intersects with First Amendment protections. For example, Stuart M. Benjamin asserts that algorithmically curated content qualifies as protected speech because algorithms are expressions of exhibit editorial control,[41] while Jimmy Fraley argues that platforms are "nothing like newspapers and television channels."[42] Paul Domer contends that because of their size and power, social media platforms should be considered public forums and thus subject to First and Fourteenth Amendment claims of violating free speech rights.[43] Jack M. Balkin calls for reimagining free expression frameworks entirely in light of platform-driven amplification.[44] Similarly, Sofia Granaki argues that platforms' content delivery algorithms can be regulated without violating the First Amendment.[45]

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[36] Arvind Narayanan, *Understanding Social Media Recommendation Algorithms*, Knight First Amend. Inst. at Columbia Univ. (2023), https://knightcolumbia.org/content/understanding-social-media-recommendation-algorithms.
[37] Arvind Narayanan, *TikTok's Secret Sauce*, Knight First Amend. Inst. (2022), https://knightcolumbia.org/blog/tiktoks-secret-sauce.
[38] Arvind Narayanan, *Twitter Showed Us Its Algorithm. What Does It Tell Us?*, Knight First Amend. Inst. (2023), https://knightcolumbia.org/blog/twitter-showed-us-its-algorithm-what-does-it-tell-us.
[39] Luca Belli & Marlena Wisniak, *What's in an Algorithm? Empowering Users Through Nutrition Labels for Social Media Recommender Systems*, Knight First Amend. Inst. (Aug. 2023), https://knightcolumbia.org/content/whats-in-an-algorithm-empowering-users-through-nutrition-labels-for-social-media-recommender-systems.
[40] Jonathan Stray, Ravi Iyer & Helena P. Larrauri, *The Algorithmic Management of Polarization and Violence on Social Media*, Knight First Amend. Inst. (Aug. 2023), https://knightcolumbia.org/content/the-algorithmic-management-of-polarization-and-violence-on-social-media.
[41] Stuart M. Benjamin, *Algorithms and Speech*, 161 U. Pa. L. Rev. 1445 (2013).
[42] Jimmy Fraley, *Social Media Platforms as Publishers: Evaluating the First Amendment Basis for Content Moderation*, 3 Princeton Legal J., no. 2 (Spring 2023).
[43] Paul Domer, De Facto State: Social Media Networks and the First Amendment, 95 Notre Dame L. Rev. 893 (2020).
[44] Jack M. Balkin, *Free Speech in the Algorithmic Society: Big Data, Private Governance, and New School Speech Regulation*, 51 UC Davis L. Rev. 1149 (2018).
[45] Sofia Grafanaki, *Platforms, the First Amendment and Online Speech: Regulating the Filters*, 39 Pace L. Rev. 1 (2018).

Legal scholars disagree about applying the concept of common carriage to social media platforms; the basis of these arguments lies in applying historical analogies.[46,47] However, these legal analyses do not go beyond cursory technical analysis in making their arguments. This may be due to a lack of digestible literature around recommender systems for non-technical researchers, apart from Narayanan—though his work does not include legal analysis.

Much of the recent legal debate surrounding recommender systems focuses on Section 230. Max Del Real contends that recommendation algorithms may not be eligible for broad immunity under Section 230 because they actively shape what content users see.[48] E. Magrani and P.G.F da Silva highlight the ethical and legal concerns raised by AI-driven recommender systems, such as opacity, discrimination, and lack of user control, but do not provide a thorough technical analysis of how these systems operate.[49] Similarly, Michael Daly Hawkins and Matthew J. Stanford critique the expansiveness of Section 230 immunity for platforms and call for a modernization (whether reinterpretation of amendment) of the legislation, but do not explore the liability of the technical implementation.[50] The Congressional Research Service has issued several reports outlining similar concerns, noting that the distinction between hosting and promoting content is increasingly blurred in recommender systems.[51,52]

Taken together, all these works establish a robust foundation for analyzing recommender systems through both technical and legal lenses. Yet very few attempt to do both at once. Most technical papers lack legal analysis, while most legal critiques assume a high-level understanding of how recommender systems work without examining their actual mechanics. This thesis seeks

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[46] Christopher S. Yoo, *What's in a Name?: Common Carriage, Social Media, and the First Amendment*, 119 Nw. U. L. Rev. 49 (2024).
[47] Morgan Ricks & Ganesh Sitaraman, *What Centuries of Common Law Can Teach Us About Regulating Social Media*, L. & Pol. Econ. Project (2024), https://lpeproject.org/blog/what-centuries-of-common-law-can-teach-us-about-regulating-social-media/.
[48] Max Del Real, *Breaking Algorithmic Immunity: Why Section 230 Immunity May Not Extend to Recommendation Algorithms*, 99 Wash. L. Rev. 1 (2024)
[49] E. Magrani & P.G.F. da Silva, *The Ethical and Legal Challenges of Recommender Systems Driven by Artificial Intelligence*, in *Multidisciplinary Perspectives on Artificial Intelligence and the Law* 133 (H. Sousa Antunes et al. eds., Springer 2024), https://doi.org/10.1007/978-3-031-41264-6_8.
[50] Michael Daly Hawkins & Matthew J. Stanford, *Uproot or Upgrade? Revisiting Section 230 Immunity in the Digital Age*, U. Chi. L. Rev. (n.d.), https://lawreview.uchicago.edu/online-archive/uproot-or-upgrade-revisiting-section-230-immunity-digital-age.
[51] Valerie C. Brannon & Eric N. Holmes, *Section 230: An Overview*, Cong. Rsch. Serv. (2024).
[52] Eric N. Holmes, *Liability for Algorithmic Recommendations*, Cong. Rsch. Serv. (2023).

to bridge that gap, arguing that meaningful regulation of recommender systems must be grounded in a detailed understanding of their internal implementation.

Chapter 3: The Technical Representation of Content

Recommender systems rely on their ability to extract and interpret data about user behavior and content. They feed this data into machine learning models, which use it for classification and ranking, ultimately outputting content recommendations. Ranking algorithms and model architecture do the "recommending" tasks, but they could not function accurately or efficiently without extensive data processing. The data processing pipeline shapes raw data into input for the models. It is important to understand how this data processing works, because it is the foundation of recommender systems.

Platforms organize data into signals during data collection. Signals represent measurable attributes of the content and user behavior; they are "raw" because they have not been processed. To improve efficiency of machine learning models, signals are structured into features, which these models use to make predictions. Engineers use various processes to collect signals, build features, and weight those features to best fit their overall goals for the system. Even though content processing does not implement actual content recommendation, it influences the output of the systems because it acts as the lens through which content is evaluated. This chapter explores signals and features while the next chapter explains the ranking and recommending functionalities.

Signals

Social media platforms use a plethora of signals to represent any single piece of content. Signals are data points that capture measurable attributes of content and user interactions, such as the binary presence of a like. In the way different people engage with the world differently because of what they chose to see, signal choice and weight fundamentally shapes the behavior of recommender systems.

Signals represent a range of quantitative and qualitative features. Quantitative signals include likes, shares, comments, and watch time (which reflects how long a user looks at a video or post) and these signals vary in format. For example, the presence of a like is a binary signal (1 for a like, 0 for no interaction) that theoretically assumes positive interest. On the other hand, a signal like watch time is continuous and recorded in seconds. Comments carry multiple dimensions of information and thus may be represented by many signals, such as length, sentiment score, presence of links, etc. A repost or share can be binary, continuous, or categorical depending on

what is measured. If a user attaches a message to a repost, textual analysis provides signals about both the post being shared and the user who shares it. Because the technical details of social media recommender systems are proprietary, it is impossible to know the complete breadth of signals.[53]

Signals can be organized into two categories: those that measure user interactions with content (engagement), and those that capture information about content itself. Later in the process, signals are aggregated together to make features. Some features may reflect strictly engagement-based or content-based signals, but others may be made up of a mix of both categories.

Explicit signals such as likes are deliberate actions that reflect the user's conscious approval of a piece of content. Implicit signals, such as watch time or scrolling behavior, are captured passively without user input. Both implicit and explicit signals quantify subjective qualities of content such as "relevance" and "value."[54] Implicit signals are empirical observations rather than conscious user expressions of preference. They represent the user's behavior, not necessarily their desires. Particularly as theories about social media addiction develop,[55] it is important to keep in mind that "interest" implies positive curiosity or enjoyment, which is not always the case for compulsive social media users.[56] Even neutral-seeming signals, like watch time, are based on subjective assumptions about what kind of behavior indicates relevance or value.

Engagement-based signals are particularly valuable to platforms because they are immediate and straightforward to measure. Examples of engagement-based signals include "How many times users watched at least *n*-seconds of a piece of content anywhere on the platform," and "How much time people have spent viewing the content in total."[57] These can be used in real-time to update recommendations. For instance, when a user likes or shares a post, that interaction directly informs the algorithm as to the perceived relevance of the post. This measurability makes engagement-based signals ideal for social media feeds, which rely on accelerated cycles of

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[53] *Chapter 6: The Commercial Unfairness of Recommender Systems on Social Media*, in *Artificial Intelligence and the Media: Reconsidering Rights and Responsibilities* 148 (Taina Pihlajarinne & Anette Alén-Savikko eds., Edward Elgar Publ'g 2022).
[54] Meta, *Instagram Reels Chaining AI System*, Transparency Ctr. (Dec. 13, 2024), https://transparency.meta.com/features/explaining-ranking/ig-reels-chaining.
[55] Alfonso Pellegrino et al., *Research Trends in Social Media Addiction and Problematic Social Media Use: A Bibliometric Analysis*, 13 Frontiers Psychiatry (2022), https://doi.org/10.3389/fpsyt.2022.1017506.
[56] Eli Pariser, *The Filter Bubble: How the New Personalized Web Is Changing What We Read and How We Think* (Penguin Publ'g Grp. 2012).
[57] Meta, *Instagram Reels Chaining AI System*, Transparency Ctr. (Dec. 13, 2024), https://transparency.meta.com/features/explaining-ranking/ig-reels-chaining.

feedback and calibration to generate engagement. However, signal values can vary as quickly as a user's mood changes, weakening their ability to consistently evaluate content quality.

In contrast, content-based signals represent characteristics of the content independent from user engagement. These signals represent data like hashtags, keywords, and subject. A video of a kitten might be associated with tags such as "animal," "cat," and "furry." Content-based signals can also capture the presence and nature of certain audio or visual elements, such as the color of the kitten's fur or where the kitten is. A happy kitten in a field of flowers might be tagged differently than a malnourished kitten in an alley.

Content-based signals require more complicated computational methods of analysis than engagement-based signals. Two examples of such methods are computer vision, which interprets images and video frames, and natural language processing, which can analyze text descriptions, captions, or comments. Interpreting content-based signals allows platforms to go beyond surface-level interactions and better analyze the content being consumed.

Some engagement-based signals do require computational analysis, such as finding comment sentiment on a post. However, there are many engagement-based signals that do not require computational analysis, whereas content-based signals require such analysis by definition. Using content-based signals comes at a higher cost, especially with the vast scale of social media platforms.

Raw data is unstructured and highly dimensional, meaning it involves many variables. Using many variables enables a more robust representation of content in theory. If you are only evaluating a pizza based on its shape, you will only see a circle—not even a piece of food. As you add more features, you begin to perceive more: its toppings, crust quality, temperature, the acidity of its sauce. However, taking in many details might ultimately inhibit your opinion and enjoyment of the pizza. After all, it is a pizza, and you are not a computer. Yet computers have a similar problem. In addition to the costliness of collecting infinite signals there is also associated computational risk. Platforms are forced to make trade-offs between computational efficiency and a richer understanding of content and behavior.

This trade-off is often attributed to a phenomenon known as the "curse of dimensionality," which describes the complications that arise as the number of input variables (or "dimensions") increases. High-dimensional data is typically sparse, meaning that many feature values are zero because any given data point typically only exhibits a small subset of the possible variables. In the

same way an individual person possesses only a limited number of traits out of all imaginable traits, a user or content item may interact with or possess only a few features from a vast variable space. The rest of the variable values are often zero, representing the absence of those traits or interactions. As dimensionality increases, distance metrics degrade in meaning, training data becomes less representative, and overfitting, where models "memorize" specific noise patterns in the training data rather than learning robust general features, becomes more likely.[58] Overfitting is analogous to a person memorizing answers off a practice test, then being stumped on the actual exam because they are unable to apply the concepts in real time

Overfitting is particularly risky for models when neural network capacity is high and the input data is noisy, both of which tend to be true with social media recommender systems.[59,60] Sparsity increases as a function of dimensionality and is addressed by using larger and larger training sets. In other words, the amount of data needed to make such predictions increases significantly in high-dimensional spaces.[61] Collecting the necessary amount of data can be very expensive.[62] Additionally, finding similarity between points becomes more difficult as distance metrics become meaningless due to concentration of measure.

However, modern machine learning does not only suffer from high dimensionality—it can also benefit. In fact, the term "blessing of dimensionality" is sometimes used in practice. An idea called the "manifold hypothesis" suggests that high-dimensional data often lies on a lower-dimensional manifold (a manifold is a curved shape that lies within a space and appears flat locally), meaning complex data can be effectively learned using fewer features.[63] Deep neural networks exploit this quality to progressively reinforce relevant features during automatic feature extraction in training. They can do this due to their layered structure and ability to minimize

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[58] Sandeep Jain, *Curse of Dimensionality in Machine Learning*, GeeksforGeeks (2024), https://www.geeksforgeeks.org/curse-of-dimensionality-in-machine-learning/.

[59] E. Aarthi et al., *Integrating Neural Network-Driven Customization, Scalability, and Cloud Computing for Enhanced Accuracy and Responsiveness for Social Network Modelling*, 14 Soc. Network Analysis & Mining, no. 1, art. 139 (Dec. 2024), https://doi.org/10.1007/s13278-024-01302-0.

[60] Leonardo Petrini et al., *Learning Sparse Features Can Lead to Overfitting in Neural Networks*, 2023 J. Stat. Mech. Theory & Experiment 11 (Nov. 2023), https://doi.org/10.1088/1742-5468/ad01b9.

[61] Essam Debie & Kamran Shafi, *Implications of the Curse of Dimensionality for Supervised Learning Classifier Systems: Theoretical and Empirical Analyses*, 22 Pattern Analysis & Applications 519 (2019), https://doi.org/10.1007/s10044-017-0649-0.

[62] Francesco Ventura et al., *Expand Your Training Limits! Generating Training Data for ML-Based Data Management*, in *Proceedings of the 2021 Int'l Conf. on Mgmt. of Data (SIGMOD '21)* 1865 (2021), https://doi.org/10.1145/3448016.3457286.

[63] Nick Whiteley and Annie Gray and Patrick Rubin-Delanchy, *Statistical exploration of the Manifold Hypothesis,* Cornell University ArXiv, (Mar. 2025), https://doi.org/10.48550/arXiv.2208.11665.

prediction error through weight adjustments. Additionally, locality (the idea that nearby data points share structure) and symmetry (invariance to input transformations) help neural networks reduce the need to explore the entire high-dimensional space. Furthermore, high concentration of measure can sometimes actually be a benefit rather than a hindrance. For example, in high dimensions even exponentially many random points are likely to be linearly separable. Each point becomes a vertex of the convex hull with high probability, making simple classifiers relatively powerful.

However, the curse is still a curse in settings where data lack structure, manifold assumptions break down, or distance metrics become unreliable. These challenges are particularly acute in social media data, which is often noisy, unstructured, and shaped by complex human behavior. Thus, the curse of dimensionality is a complication for social media platforms because it forces them to limit the number of input variables into their machine learning models. In response, engineers create feature vectors (features), which are in effect a processed, compressed function of multiple signals. Features condense information from a high-dimensional signal space into lower-dimensional inputs. Thus, when creating features platforms are faced with a trade-off: Incorporating more signals can capture more nuance at the cost of accuracy and efficiency, while relying on fewer signals yields more basic recommendations. Model performance is heavily dependent on the representation of the feature vectors; as a result, a huge part of deploying machine learning models is extracting, selecting, and engineering effective features.[64]

Features

Because signals are unprocessed, they often contain irrelevant information (called noise) that can obscure meaningful patterns in the data. Raw data is also highly dimensional and unstructured. For machine learning (ML) models to perform effectively and efficiently, signals must be transformed into cleaner, more informative inputs called features.

Like signals, there are various types of features. Features can be user-based, item-based, interaction-based, or contextual. They can also be multimodal aggregations of signals.[65] Hybrid approaches combining different types of features can be effective for capturing complex patterns

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[64] J. Heaton, *An Empirical Analysis of Feature Engineering for Predictive Modeling*, in *Proceedings of the 2016 IEEE SoutheastCon* 1 (2016), https://doi.org/10.1109/SECON.2016.7506650.

[65] Qidong Liu et al., *Multimodal Recommender Systems: A Survey*, 57 ACM, no. 2 (Sept. 2024), https://doi.org/10.1145/3695461.

in social media data.[66] Engineers employ a process called feature extraction to transform raw signals into features. In some cases where automatic feature extraction is not possible, platforms could theoretically use manual annotation. However, wide manual annotation is not sustainable because of cost.[67] In recommender systems, these extracted features are fed into ML models which rank and recommend content based on predicted user preferences.

Feature extraction transforms raw data into processable features while preserving structure and patterns. Social media content generates a huge variety of signals including numerical, textual, audio, and visual information. The feature extraction process transforms these raw signals into structured features which capture information about content and user behavior. For example, text data can be preprocessed to remove punctuation and stopwords, then converted into numerical vectors using techniques like Term Frequency-Inverse Document Frequency (TF-IDF) and word embeddings.[68] Embeddings are numerical representations of data (such as users or items) in lower-dimensional space, thus expressing complex inputs as vectors or numbers that machine learning models can understand.

The number of features that may be used efficiently by an ML model is finite, so platforms must determine which features to use. Not all features have the same predictive power. For example, a signal like "Whether or not you've been viewing content for more than *n*-seconds" might be more significant than "How many times the content was dismissed." Feature importance quantifies the contribution of a given feature to the effectiveness of a model; feature selection finds and selects features with high importance.[69] Selecting and weighting features properly is an important and challenging task for machine learning engineers.

Techniques of feature selection are diverse and context specific. Traditional approaches rank features based on importance determined by statistical models such as correlation. For instance, Recursive Feature Elimination (RFE) is a method that trains a model to rank all features

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[66] S. Singh, K. Kumar & B. Kumar, *Sentiment Analysis of Twitter Data Using TF-IDF and Machine Learning Techniques*, in *Proceedings of the 2022 Int'l Conf. on Machine Learning, Big Data, Cloud & Parallel Computing (COM-IT-CON)* 252 (2022), https://doi.org/10.1109/COM-IT-CON54601.2022.9850477.

[67] Dietmar Jannach, Markus Zanker & Gerhard Friedrich, *Recommender Systems: An Introduction* 76 (Cambridge Univ. Press 2011).

[68] Farma Elsafaoury, Steven R. Wilson & Naeem Ramzan, *A Comparative Study on Word Embeddings in Social NLP Tasks*, in *Proceedings of the 10th Int'l Workshop on Natural Language Processing for Social Media (SocialNLP 2022)* 44 (July 2022), https://doi.org/10.18653/v1/2022.socialnlp-1.5.

[69] Timothy R. Brick et al., *Feature Selection Methods for Optimal Design of Studies for Developmental Inquiry*, 73 J. Gerontology Series B: Psychol. Sci. & Soc. Sci. 113 (Feb. 2017), https://doi.org/10.1093/geronb/gbx008.

by importance, removing the least important features after each training iteration and continuing until the optimal subset of features is found. The goal of RFE is to reduce a large set of features to a smaller subset of the most prominent variables before inputting them into a machine learning model.[70] The Boruta algorithm is built around the random forest classification algorithm. Boruta assesses the significance of variables by adding 'shadow' attributes, which are shuffled versions of the real features, and compares the performance of real feature against the best-performing shadow attributes. The algorithm re-shuffles the features into new shadow attributes each run. If a feature consistently outperforms the shadow attributes, it is marked as important.[71] Boruta seeks to find all relevant features, not just a minimal subset like RFE.[72]

Conventional feature selection methods typically assume that data instances are independently and identically distributed (i.i.d.), but in reality, social media data points are inherently linked to each other. To account for the graphical (connected) nature of the data, advanced feature selection frameworks incorporate social relationships.[73] In these cases, feature selection depends on several factors such as the availability of labels, structure of the data, the ranking algorithms, and the specific goals of the platform. For instance, when engagement signals are abundant, but user feedback is sparse, semi-supervised or unsupervised methods of feature selection can be used.[74]

Effective feature selection in social media recommender systems require platforms to find a balance between computational efficiency, predictive power, and adaptability to dynamic user behavior. Engineers constantly update feature sets as platform goals shift, user preferences change, and new interaction patterns develop. The moving target is to improve model accuracy while simultaneously aligning the system's recommendations with the broader strategic objectives of the platform.

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[70] Brick et al., *supra* note 69.
[71] Miron B. Kursa and Witold R. Rudnicki, *Feature Selection with the Boruta Package*, 36 J. Stat. Software 11 (September 2010), https://doi.org/10.18637/jss.v036.i11.
[72] Rung-Ching Chen et al., *Selecting Critical Features for Data Classification Based on Machine Learning Methods*, 7 J. Big Data 52 (July 2020), https://doi.org/10.1186/s40537-020-00327-4.
[73] Jiliang Tang & Huan Liu, *Feature Selection for Social Media Data*, 8 ACM Transactions on Knowledge Discovery from Data, no. 4, art. 19 (Oct. 2014), https://doi.org/10.1145/2629587.
[74] Jiliang Tang & Huan Liu, *Unsupervised Feature Selection for Linked Social Media Data*, in *Proceedings of the 18th ACM SIGKDD Int'l Conf. on Knowledge Discovery & Data Mining* 904 (Aug. 2012), https://doi.org/10.1145/2339530.2339673.

Feature selection for social media platforms involves constant reevaluation. Engineers monitor performance metrics relative to platform goals and adjust feature selection methods accordingly. They can introduce new signals as user behavior evolves to optimize those metrics. If a platform aims to maximize engagement, it will prioritize whatever features result in the highest levels of engagement. If it aims to increase quality discourse in its comment section, it might prioritize whatever signals keep polarization metrics low. Thus, feature selection is not just about computational efficiency; it is also a series of decisions about what aspects of content and behavior a platform deems most relevant for shaping user experience.

Engagement data such as watch time and click-through rate can be aggregated to create more holistic features that reflect user interest and behavior. For example, a feature representing average watch time per session captures more nuanced patterns of user engagement than a singular piece of watch time data. Platforms create composite features that combine engagement data with contextual signals to adjust recommendations based on specific user context. These aggregated composite features can be fed into reinforcement learning models, where user behavior is treated as a feedback loop.

Feature engineering extends beyond engagement data to include content-based signals. For instance, embeddings from a language model like BERT (Bidirectional Encoder Representations from Transformers) encodes semantic similarity between comments or posts.[75] BERT can be combined with user behavior data to capture not only what the user likes, but who they are i.e. their personal identity. Feature engineering requires a balance between static features (like content metadata) and dynamic features (like real-time engagement patterns). Engineers continuously determine which features are prioritized to meet the platform's overall goals. The process is iterative, and the definition of "relevant features" evolves as user behavior and platform objectives do so themselves.

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If a platform aims to maximize user engagement on social media, it will prioritize the features which will generate increased engagement further on in the recommendation pipeline.

[75] Mohamad Merchant, *Semantic Similarity with BERT*, Keras (2020), https://keras.io/examples/nlp/semantic_similarity_with_bert/.

Instagram Reels optimizes for engagement by using signals like watch time, comments, and shares to build features.[76] Its recommender system is designed to suggest content that will keep users scrolling for an extended period. Compare that with Netflix, which operates a subscription-based model instead of an advertising-based one and optimizes for long-term user satisfaction and retention. Its algorithm may prioritize features like viewing history and genre preference. Unlike Reels where engagement is the primary goal, Netflix focuses on keeping subscribers. Varying goals and implementations demonstrate the subjectivity of the content processing pipeline in recommender systems.

When interpreting human preference through data analysis, there appear to be infinite avenues for subjectivity. As an example, let's use preference towards peanuts to illustrate this point. American peanut popularity is a cultural phenomenon that emerged as a result of violent historical ideologies and systems. If not for the North Atlantic slave trade, peanuts might not have made it to North America or become highly commercialized during the Civil War.[77] That every American has an opinion about peanuts is not a certainty, but it is probably a reasonable assumption due to popular marketing and availability. A person's reaction to eating a peanut can be harvested as signals, those signals turned into features based on what we surmise they represent, and those features weighted as parameters according to what we feel is most predictive towards their opinion. Notice how many times I used hedging language in that paragraph. Though the data processing pipeline is certainly more complex and quantitative than this simple example, the fundamental feature of assumption-making is present throughout.

Because features determine what a recommender system "sees," they inevitably shape what it predicts and suggests. When building features from signals, social media platforms look for the most efficient and effective inputs for their machine learning models. An effective recommendation is one that elicits high user engagement and retention. Platforms select, engineer, and then weight features to reflect their overall goals for the system, ultimately reinforcing specific patterns of engagement and interaction by shaping what users see. These ranking decisions, in turn, influence broader questions about the expressiveness of platforms and their role in shaping information access and discourse.

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[76] Meta, *Instagram Reels Chaining AI System*, Transparency Ctr. (Dec. 13, 2024), https://transparency.meta.com/features/explaining-ranking/ig-reels-chaining.

[77] Virginia Carolina Peanuts, *The History of Peanuts*, Va. Carolina Peanuts, https://www.aboutpeanuts.com/all-about-peanuts/origin-history-of-peanuts (last visited Mar. 20, 2025).

Chapter 4: The Recommendation Itself

Social media platforms select and engineer features which best facilitate their overall goals for machine learning model performance. They also build elaborate technical architecture through which models rank and recommend content, using continuous refinement processes to improve model performance based on certain quantifiable metrics. Like features, these metrics reflect both human and algorithmic choice.

Performance metrics are optimization targets that reflect overall platform goals as well as computational efficiency. Metrics are derived from weighted averages of engagement signals available to the platform. For example, Facebook uses a weighted average of likes, reactions, reshares, and comments they call "Meaningful Social Interactions (MSI)" which Narayanan hypothesized as MSI(user, item) = affinity(user, poster) $\times \Sigma_{\text{int-type}}\, P(\text{user, item, int-type}) \times \text{Weight[int-type]}$ where P(user, item, int-type) is the predicted probability that the user will have a specific type of interaction with the given item.[78] Engagement probability is generated by machine-learning models. As a model makes predictions, it subsequently relies on metrics as benchmarks to iteratively adapt and improve performance.

This chapter will not attempt to provide an exhaustive account of all technical layers involved in recommender systems. There are many processes that play critical roles in production environments but fall outside the scope of this thesis, such as online learning, reinforcement learning, and advanced clustering strategies. Nonetheless, the systems and methods discussed here serve as core examples that illustrate a broader theme: engineers actively shape how recommender systems behave despite extensive automation and technical complexity.

Recommender System Architecture

A machine learning (ML) model is a computational system that uses algorithms to identify patterns in data and makes predictions or decisions based on those patterns. In other words, ML models generalize from historical examples to handle unseen (new) data. Today, tasks such as classification, ranking, and recommendation are implemented by these models because they can

[78] Arvind Narayanan, *Understanding Social Media Recommendation Algorithms*, Knight First Amend. Inst. at Columbia Univ. (2023), https://knightcolumbia.org/content/understanding-social-media-recommendation-algorithms.

manage streams of data that are too large for human processing. The performance of a model means how well a model makes accurate predictions or classifications on unseen data, typically measured using metrics like accuracy, precision, recall, loss, or task-specific scores like MSI. What counts as a "correct" or "accurate" prediction is defined by engineers based on platform objectives; if a model aims to maximize engagement, its best-performing predictions are those which lead to more user interaction.

"Feed Recommendation" is one of several major social media functionalities used for content propagation. It drives a large fraction of engagement and affords platforms almost complete control over what a user sees.[79] The behavior and output of Feed Recommendation is driven by underlying algorithms. Since there are many algorithms and social media recommender systems are generally not open source, we can only speculate about which techniques platforms use.[80] The infamous "recommendation algorithm" itself is made up of many algorithms, most with proprietary protections. Briefly explaining common algorithms and their functionalities helps reveal the inner workings of the system. We may notice that a common thread throughout the system is that all technical processes rely on metrics to govern performance.

Recommender systems are layered like Russian dolls, with algorithms nested inside one another to perform specialized tasks. We will discuss two broad types of recommender systems: collaborative filtering and content-based filtering. To understand the distinction, we will return briefly to our peanut analogy from Chapter 3. A content-based filtering system would look at each person's individual preferences and characteristics. If you've enjoyed peanut butter cookies and peanut butter sandwiches in the past, the system might predict you're likely to enjoy Reece's Cups. It doesn't matter if no one else in history has ever tried Reece's Cups, because the system knows the candy has peanuts in it and therefore you might like it. By contrast, a collaborative filtering system doesn't care about peanuts specifically. This type of system looks at what other people like you have enjoyed. If you and a group of users have all rated almond butter snacks highly, and many of those other users have also rated Reece's Cups highly, the system might recommend Reece's Cups to you even if you have never tried them. Collaborative filtering looks at what people with similar preferences to yours tend to rate highly, whereas content-based filtering builds your preference profile based on your past behavior and recommends similar items. In practice,

[79] Narayanan includes Search, Ad delivery and targeting, Trending, Content moderation, Friend recommendation…
[80] Apart from Bluesky.

platforms often combine approaches,[81] but understanding the strengths and limitations of these two main types provides insight into how more complex architecture, like two-tower and multimodal models, are constructed.

Collaborative Filtering

Collaborative filtering recommends content based on patterns in user behavior, such as what users have liked or interacted with in the past. The key inputs to collaborative filtering methods are past user-item interactions (likes, shares, and watch time) and the output is a ranked list of items the system thinks a user will enjoy. Engineers can alter these systems by adjusting how similarity is calculated or by changing how much weight is given to popular versus niche items, depending on whether they want to prioritize novelty, engagement, or personalization.

There are two types of collaborative filtering: model-based and memory-based. The difference between memory-based and model-based methods lies in how they generate their recommendations. Model-based approaches first process the raw data offline to train a model. Then, at runtime, they use this pre-trained model to quickly generate predictions.[82] This makes them more scalable for large systems compared to memory-based methods, which compute recommendations in real time. Instead of relying on a stored database at runtime, model-based methods extract features from the dataset during training and use those features to inform future predictions.[83]

Memory-based methods store the rating database in memory and reference it to generate each recommendation, while model-based methods use data mining and machine learning techniques to predict the user's rating for an unseen item. Memory-based methods use a utility matrix to recommend new items by considering the preferences of its neighborhood. The utility matrix is a table that represents the preferences or interactions between users and items. Each row corresponds to a user, each column to an item, and each cell contains a value that reflects how the user interacted with the item. Memory-based filtering gets its name from the original rating database, which the system stores in memory and uses directly to generate recommendations.[84]

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[81] Deepjyoti Roy & Mala Dutta, *A Systematic Review and Research Perspective on Recommender Systems*, 9 J. Big Data 59 (2022), https://doi.org/10.1186/s40537-022-00592-5.
[82] Dietmar Jannach, Markus Zanker & Gerhard Friedrich, *Recommender Systems: An Introduction* 26 (Cambridge Univ. Press 2011).
[83] Roy and Dutta, *supra* note 81.
[84] Jannach et al., *supra* note 77.

Though memory-based methods are theoretically more precise because they use the full dataset to generate recommendations,[85] scalability is a major concern because poor scalability lowers system performance and worsen responsiveness to user requests.[86]

There is a further distinction within memory-based methods: user-based and item-based filtering.[87] In the former, the system predicts how a user will rate a new item by finding other users with similar past behavior—known as the user's neighborhood—who have already rated that item. It then calculates the predicted rating for a piece of content based on those users' ratings (i.e. their engagement data.)

The latter, item-based filtering, compares items instead of users. Rather than forming a user neighborhood, the algorithm builds an item neighborhood by grouping items that users have frequently rated similarly. The algorithm then recommends items that are most similar to those the user has already liked. Neighbor methods and matrix factorization are just two examples of algorithmic approaches to identify user and item neighborhoods.

Neighbor methods compute the distance between data points using quantitative distance metrics like cosine similarity or Euclidean distance. The k-Nearest Neighbors (kNN) algorithm is perhaps the most famous nearest neighbor method, because it is straightforward to comprehend. The algorithm works by finding the *k* closest data points (referred to as neighbors) to a given input based on the chosen distance metric. Distance depends heavily on how data is represented, thus the output of kNN is dictated by feature extraction, selection, and engineering. For example, calculating similarity based on watch time of posts may not return the same set of recommendations as using the number of comments on posts. In classification, the algorithm assigns the most common label among the neighbors to the input. For regression tasks, it takes the (sometimes weighted) average of the neighbors' values. The kNN algorithm is easy to implement but it can be slow and ineffective with large datasets or complex data, nevertheless it is foundational for recent approaches.[88] Collaborative filtering relies on nearest neighbor methods to identify user and item neighborhoods.

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[85] Jannach et al., *supra* note 82.
[86] Bahrudin Hrnjica, Denis Music & Selver Softic, *Model-Based Recommender Systems*, in EAI/Springer Innovations in Communication and Computing 125, 125–46 (2020), https://doi.org/10.1007/978-3-030-40037-8_8.
[87] Roy and Dutta, *supra* note 81.
[88] Y. Vadukiya et al., *K-Nearest Neighbor Based Friends Suggestion on Social Media*, in *Computing and Machine Learning* 13 (J.C. Bansal et al. eds., vol. 1144, Springer 2025), https://doi.org/10.1007/978-981-97-7839-3_2.

Matrix factorization and nearest neighbors differ in their implementation, but both aim to quantify the similarity between users or items. Matrix factorization relies on the assumption that sparse datasets contain various latent features, which can be generalized into categories rather than analyzing every single user and item in the dataset. This is a simple embedding model that represents two matrices: a user matrix $U \in R^{m \times d}$ where m is the number of users and each row i is a d-dimensional embedding for user i, and an item matrix $V \in R^{n \times d}$ where n is the number of items, and each row j is a d-dimensional embedding for user j. The embeddings are learned so that the product UV^T accurately approximates the feedback matrix A. The (i, j) entry of this product represents the dot product of the embeddings for user i and item j, which should closely match $A_{i,j}$.[89] The goal of matrix factorization is to train the embeddings so that their interaction captures actual engagement behavior, resulting in accurate predictions.

Collaborative filtering bases its predictions off the behavior of individual users, so it relies on engagement data. It is relatively convenient to collect this data, because engineers aren't required to manually extract and summarize the information as they would to make content-based features such as semantic similarity. User-based filtering can expose users to a broad field of information because it is based on the similarity of user behavior, and though general interests may align, every user is unique. In general, collaborative filtering can personalize quickly and improves as users respond dynamically.[90]

On the other hand, there are some considerable drawbacks to collaborative filtering. Most significantly, this method struggles with data sparsity.[91] Data sparsity refers to the phenomenon where most of the data in a dataset is missing or consists of zero values. In recommender systems, sparsity causes many blank cells in the utility matrix because users have only interacted with a tiny subset of all possible items. Since collaborative filtering algorithms rely on finding similar users, sparse data is problematic because each user has interacted with a unique set of items and therefore has a unique set of preferences. Another challenge with collaborative filtering is the cold start problem. This is the inability of systems to generalize when given a new user whose behavior and preferences are unknown. If a user has just joined a platform, the platform has no historical data

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[89] Google, *Matrix Factorization*, Google Machine Learning, https://developers.google.com/machine-learning/recommendation/collaborative/matrix (last visited Mar. 28, 2025).
[90] Xinyi Wu, *Comparison Between Collaborative Filtering and Content-Based Filtering*, 16 Highlights in Sci., Eng'g & Tech. 480 (2022), https://doi.org/10.54097/hset.v16i.2627.
[91] *Id*.

about their behavior and is thus unable to make high-quality recommendations.[92] Additionally, the utility of user-based collaborative filtering increases in settings where users' tastes are homogenous and they show deep interest in the target item,[93] which is not often the case on social media.[94]

Content-Based Filtering

While collaborative filtering makes predictions by identifying other users with similar preferences, content-based filtering focuses solely on the target user's own interaction history. Instead of comparing users, it recommends new items that share features with those the user has previously liked. The inputs include both the target user's history and content-based attributes of each item. The output is a ranked list of new items that are similar to those the user has previously engaged with. The recommendations produced by content-based filtering are relatively explainable—engineers can explain why the system is making a particular recommendation—since they are based completely on the target user's engagement history. However, this quality makes the approach prone to filter bubbles, where a user is repeatedly shown a narrow range of content aligned with their existing interests.[95] Engineers can modify the feature set, introduce diversity constraints, or add weights to certain item attributes to promote exploration or ensure exposure to a wider range of content.

Content-based systems rely on three key components: item representation, user profile, and matching and prediction. Like collaborative filtering, item representation is created during the data processing pipeline by building features from raw data signals. The user profile is created from explicit and implicit feedback from the target user and evolves as the system collects more data from that user. For matching and prediction, content-based systems compare new items to items the user has liked in the past and recommend based on relevance. As with collaborative filtering, relevance in content-based filtering remains subjective because it depends on how the content is

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[92] However, they could rely on demographical data inferred from IP address such as location and device, in addition to sign-up data such as birthday and gender.

[93] P.J. Gai & A.-K. Klesse, *Making Recommendations More Effective Through Framings: Impacts of User- Versus Item-Based Framings on Recommendation Click-Throughs*, 83 J. Marketing 61 (2019), https://doi.org/10.1177/0022242919873901.

[94] S.S. Sundar et al., *Sharing Without Clicking on News in Social Media*, 9 Nat. Hum. Behav. 156 (2025), https://doi.org/10.1038/s41562-024-02067-4.

[95] Eli Pariser, *The Filter Bubble: How the New Personalized Web Is Changing What We Read and How We Think* (Penguin Publ'g Grp. 2012).

represented in the system—even when platforms apply mathematical definitions of distance and similarity.

Various algorithms can be used to learn a function that models an individual user's interests, including decision trees, nearest neighbor methods, Rocchio's algorithm, linear classifiers, and probabilistic methods.[96] Because content-based filtering does not have to learn the preferences of other users in a system outside of the target user, it adapts well to changing user preferences.[97] For the same reason, the approach is not as vulnerable to the cold start problem and data sparsity as collaborative filtering. Additionally, content-based filtering is beneficial for groups with specific interests and a high degree of consistency because it is not an issue for them if the platform is only able to collect a small amount of engagement information about niche content.[98] This quality, while often advantageous, can also cause the filter-bubble effect since it centers around reinforcing user preference rather than introducing any new types of content.

Another challenge with content-based filtering is that it requires extensive item knowledge to make accurate recommendations. The information extracted by the algorithm is more likely to contain noise—anything that doesn't reflect a clear or useful pattern, such as activity from bot accounts—thus increasing processing difficulty. The effectiveness of content-based systems depends on the system's ability to accurately and completely understand content. If the system is unable to extract the "essential information and core content" of a post, analysis will remain superficial, and the quality of recommendations is unlikely to be high.[99] As Xinyi Wu writes, "[The] content-based filtering algorithm is not available to make hierarchical recommendations for things [whose] qualities are judged by users' subjective feelings."[100] Therefore, like for collaborative filtering, the choices platforms make about how to build features largely dictate the output of content-based filtering systems.

Recommender systems have evolved to combine collaborative and content-based filtering to maximize the benefits and minimize the problems of each.[101] This hybrid architecture is still

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[96] M.J. Pazzani & D. Billsus, *Content-Based Recommendation Systems*, in *The Adaptive Web* 325 (P. Brusilovsky et al. eds., vol. 4321, Springer 2007), https://doi.org/10.1007/978-3-540-72079-9_10.
[97] Roy and Dutta, *supra* note 81.
[98] Wu, *supra* note 90.
[99] *Id.*
[100] *Id.*
[101] Shaoyun Shi, Min Zhang, Yiqun Liu & Shaoping Ma, *Attention-Based Adaptive Model to Unify Warm and Cold Starts Recommendation*, in *Proceedings of the 27th ACM Int'l Conf. on Information & Knowledge Mgmt. (CIKM '18)* (Oct. 2018), https://doi.org/10.1145/3269206.3271710.

influenced by the makeup of features. Collaborative filtering is affected by engineering choices about what constitutes "interest," while content-based filtering is affected by choices about what someone might be paying attention to in content. Though hybrid approaches are increasingly effective at achieving engineer goals, for the most part they do not render these choices unnecessary or objective, which we see through examples of advanced architecture.

Advanced Architecture

In recent years, recommender systems have advanced to incorporate more sophisticated architecture designed to capture a high level of complexity and nuance.[102] We will discuss three major advanced architectural approaches: two-tower, multimodal, and deep learning. A two-tower architecture represents user and items separately and find matches based on similarity in an abstract vector space. Multimodal systems incorporate varied data types—such as text, audio, and images—to enrich preference understanding. Deep learning can detect complex patterns in raw data thereby escaping the need for feature selection and engineering, but recommendations can be difficult to explain. For each approach, engineers can tweak architecture, change how user/item embeddings are trained, or fine-tune which types of data are emphasized to prioritize different objectives like engagement, retention, and fairness.

Advanced architecture addresses challenges such as data sparsity, cold-start, and dynamicity of user behavior. Recommender systems may use many advanced architectural approaches coupled together. Like data processing, there are layers and layers of technical processes. Though differing in approach and implementation, the behavior of any advanced architecture largely depends on the quality of its input and its performance metrics.

Two-Tower Systems

Two-tower recommender systems are used widely to handle real-time recommendation tasks in large online platforms.[103] They rely on a dual-encoder architecture which uses two separate encoders, or "towers," to independently process user and item data. The user tower learns embeddings based on user features, while the item tower learns embeddings based on item

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[102] Y. Su, Y. Li & Z. Zhang, *Two-Tower Structure Recommendation Method Fusing Multi-Source Data*, 14 Electronics 1003 (2025), https://doi.org/10.3390/electronics14051003.

[103] Amit Kumar Jaiswal, *Towards a Theoretical Understanding of Two-Stage Recommender Systems*, arXiv (2024), https://doi.org/10.48550/arXiv.2403.00802.

features. The outputs of both towers are then combined, typically using a dot product or cosine similarity, to generate a relevance score, which ranks the items for recommendation. Various algorithms can be used in two-tower systems to handle different issues. For instance, selection bias occurs when the data used to train or evaluate a model is not representative of the broader population of behavior it is supposed to reflect. Negative sampling is a technique used to help ML models learn what a user doesn't like by showing the model unseen items. When the model chooses which items to use as these unseen examples, or "negatives," it can miss important patterns. This in turn leads to skewed results and inaccurate predictions. Techniques such as Mixed Negative Sampling reduce the selection bias from negative sampling by ensuring that all items in the corpus have a chance to serve as negatives.[104] This helps the retrieval model include fresh and long-tail items because it will not automatically exclude items that lack historical user feedback.

Another approach to two-tower systems, the Adaptive-Mimic Mechanism (AMM), offers an alternative to traditional two-tower systems by improving interaction modeling and item representations. AMM does this by incorporating information between the user and item towers. The first step in AMM is the embedding layer, where features from the user and item matrices are transformed to low-dimensional vectors. Next, the system concatenates the query and candidate item vectors with their corresponding feature embeddings to create augmented input vectors for both towers. These augmented input vectors contain not only information about the current query and item, but also information from historical interactions.[105] Note that the choice of features in the embedding layer directly influences what information the augmented input vectors include, and ultimately what patterns the model can learn.

Multimodal systems

Recommender systems rely on engagement data to make accurate recommendations. However, engagement data tends to be sparse, which poses problems for model estimation and generalization.[106] To address the issue of data sparsity, multimodal recommender systems leverage auxiliary data, which is "information beyond user-item interactions, which can hold clues on how

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[104] Ji Yang et al., *Mixed Negative Sampling for Learning Two-Tower Neural Networks in Recommendations*, in *WWW '20: Companion Procs. of the Web Conf. 2020* (Apr. 2020), https://doi.org/10.1145/3366424.3386195.
[105] Yantao Yu et al., *A Dual Augmented Two-Tower Model for Online Large-Scale Recommendation*, in *Procs. of DLP-KDD 2021*, https://doi.org/10.1145/1122445.1122456.
[106] *See* Chapter 3.

users consume items" to improve recommendations.[107] Various types of auxiliary data, also known as modalities, are processed with special encoders to generate embeddings. There are different encoders for different modalities, for instance Convolutional Neural Networks (CNNs) for images and Bidirectional Encoder Representations from Transformers (BERT) for text. Then, those embeddings are fused together. The attention mechanism, which identifies and weights features of the input according to their relevance to the given task, is widely used for this fusion.[108] Attention can be personalized to each user. For instance, when considering user feedback expressed through comments, the attention mechanism might identify the most important words in comments for each user/item rather than using the same importance weights across the entire population.[109] After applying attention, the fused representation is used to predict user preference scores.

While user behavior provides the raw data for multimodal systems, the modality selection and implementations of encoder and attention ultimately determine the behavior of the system. If a platform determines post relevance by looking at sentiment of comments and image quality, it will recommend different content than one that looks at sentiment of comments and audio quality.

Deep Learning

Deep learning-based systems are based on the concept of the artificial neural network (ANN),[110] which is an architecture based on the biological processes of the same name. In computer-based neural networks, interconnected units called neurons are organized into layers. Each neuron receives an input and applies a mathematical transformation before passing the result to the next layer. These neural networks learn complex patterns through this layered structure, making deep learning systems capable of learning complex, dynamic, non-linear relationships

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[107] Aghiles Salah, Quoc-Tuan Truong & Hady W. Lauw, *Cornac: A Comparative Framework for Multimodal Recommender Systems*, 21 J. Mach. Learning Rsch. 1 (2020), https://ink.library.smu.edu.sg/cgi/viewcontent.cgi?article=6952&context=sis_research.

[108] Qidong Liu et al., *Multimodal Recommender Systems: A Survey*, 57 ACM, no. 2 (Sept. 2024), https://doi.org/10.1145/3695461.

[109] Hongtao Liu et al., *NRPA: Neural Recommendation with Personalized Attention,* in *Proceedings of the 42nd Int'l ACM SIGIR Conference on Research & Development in Information Retrieval* 1233 *(2019),* https://doi.org/10.1145/3331184.3331371.

[110] Iqbal H. Sarkar, *Deep Learning: A Comprehensive Overview on Techniques, Taxonomy, Applications and Research Directions,* 2 Sn Comput. Sci. 420 (2021), https://doi.org/10.1007/s42979-021-00815-1.

between user behavior, content attributes, and contextual data.[111] In practice, deep learning likely dominates the world of recommender systems because of these capabilities.[112]

Unlike batch-trained models which train all at once using a fixed dataset, deep learning models update themselves continuously as they access new data. On social media, they are able to adjust recommendations using real-time data, which ensures that recommendations remain relevant even as user preference evolves.[113] There are various qualities and flavors of deep learning which address different challenges in recommender systems, including sequence models for historical behavior, multimodal information fusion to address sparseness, multi-task learning for optimizing various metrics, and reinforcement learning to consider short-term and long-term benefits of the system.[114]

A deep learning system for social media might involve two neural networks: one for candidate generation and the other for ranking.[115] In candidate generation, the network selects a small set of items based on broad user profiles. The ranking network then assigns scores to each candidate based on detailed user-item interactions. In this pipeline, candidate generation is optimized for recall by focusing on retrieving relevant content from a large corpus and the ranking model is optimized for relevance by scoring and ordering those retrieved candidates.[116]

While traditional recommender algorithms such as collaborative filtering and matrix factorization are limited by the assumption that the relationship between input and output is linear, deep learning models avoid this limitation by using neural networks to learn more abstract, hierarchical feature representations directly from raw data. Multimodal systems often rely on deep learning by using techniques like convolutional neural networks and multi-layer perceptrons. Models such as Neural Collaborative Filtering and Deep Factorization Machines have demonstrated strong performance by combining deep learning's feature extraction capabilities

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[111] Quang-Vinh Dang & Claudia-Lavinia Ignat, *dTrust: A Simple Deep Learning Approach for Social Recommendation*, in *Proceedings of the 3d IEEE Int'l Conf. on Collaboration & Internet Computing (CIC)* 209 (2017), https://doi.org/10.1109/CIC.2017.00036.

[112] Ling Huang, *Recommendation System — Ranking Models*, Medium (Dec. 2023), https://medium.com/@linghuang_76674/recommendation-system-ranking-models-916efb78e3dc.

[113] Taesup Moon et al., *Online Learning for Recency Search Ranking Using Real-Time User Feedback*, in *Proceedings of the 19th ACM Int'l Conf. on Information & Knowledge Mgmt. (CIKM '10)* 1501 (Oct. 2010), https://doi.org/10.1145/1871437.1871657.

[114] Huang *supra* note 112.

[115] Paul Covington, Jay Adams & Emre Sargin, *Deep Neural Networks for YouTube Recommendations*, in *Proceedings of the 10th ACM Conference on Recommender Systems (RecSys '16)* 191 (2016), https://doi.org/10.1145/2959100.2959190.

[116] *Id*.

with collaborative filtering's ability to model user-item relationships.[117] That deep learning enables platforms to infer preference despite data sparsity and high-dimensionality makes them well-suited for social media platforms.

Though they are highly flexible and scalable, deep learning models require thorough and careful training and maintenance to achieve optimal performance. Furthermore, because they learn abstract representations of users and content that don't correspond directly to human analysis, it is hard to know why a deep learning model made a particular recommendation. While a traditional content-based filtering system might say "we're recommending this makeup tutorial because you like makeup tutorials," a deep learning model might generate a recommendation based on latent high-dimensional patterns in the data like subtle patterns in user viewing sequences. In this way, deep learning models operate as black boxes: They receive inputs, process them through layers of weights and activations, and produce an output without providing a straightforward rationale. This lack of transparency is challenging for accountability and debugging. Though there are techniques to facilitate interpretability, "they rarely form a substantial part of machine learning workflows and pipelines."[118]

Two-tower, multimodal, and deep learning systems are three prominent examples of advanced architecture for highly personalized algorithmic curation. In general, the technical objective of each architecture is to learn patterns from data through feedback and iteration, but implementation depends on platform objectives, the nature of the available data, and the computational resources available. The accuracy and efficiency of these processes are of the utmost importance for social media platforms who want their recommendations to generate high engagement and require minimal computing cost.

Model Parameters

While features come from raw input data, parameters are learned weights that define how much each feature influences the model's prediction. The model starts with randomized or initialized parameters and adjusts those parameters during training to minimize the error between

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[117] Shuai Zhang, Lina Yao, Aixin Sun & Yi Tay, *Deep Learning Based Recommender System: A Survey and New Perspectives*, 1 ACM Computing Surv., no. 1 (July 2018), https://doi.org/10.1145/3285029.
[118] Pantelis Linardatos, Vasilis Papastefanopoulos, & Sotiris Kotsiantis, *Explainable AI: A Review of Machine Learning Interpretability Methods,* 23 Entropy 18 (2021), https://doi.org/10.3390/e23010018.

predictions and actual behavior. Model parameters are essential because they fine-tune how the model interprets input; because features are not pre-weighted, the model must learn which to prioritize based on patterns in the data. Parameter values are learned by the model over iterations of training, and those parameters ultimately define how much each feature influences prediction.

Model parameters vary depending on architecture. For example, a linear regression model uses parameters as weights assigned to each feature, and the weighted sum determines the final prediction. In neural networks, each connection between neurons has a weight that determines how much influence the input from one neuron has on the next. Neural networks also use biases as parameters, which are added to the weighted sum of inputs in each neuron. For large-scale recommender systems, parameters may include latent vectors that represent users and items in a shared embedding space. This allows the model to infer similarities and differences even when feedback data is sparse. The sheer number of parameters in these systems means that training involves optimizing millions of values and often relies on sophisticated techniques like stochastic gradient descent and regularization to prevent overfitting and ensure effective generalization.

Hyperparameter Tuning

We will return again to our peanut analogy from Chapter 3. Different people require different "settings" to function well with regard to peanuts. Some people are severely allergic, some just don't like them, and others eat them by the handful daily. If we were building a recommender system for peanuts, the features might include how often a person eats peanuts, whether they have a peanut allergy, or how they rate peanut-based snacks. The model parameters would be the internal values learned during training—for example, the model might learn that a person who buys a jar of peanut butter every week has a strong positive preference, while someone with an allergy should never be recommended a peanut product.

Hyperparameters are different from features or parameters in that they are the settings that govern how the model learns from the data, and they control model structure, function, and performance.[119] In our peanut analogy, hyperparameters would be like adjusting how quickly the model updates when learning from new data about peanut preferences. If the learning rate is too high, the model might be overly sensitive to irrelevant information, assuming a person loves

[119] AWS, *Is Hyperparameter Tuning?,* AWS, https://aws.amazon.com/what-is/hyperparameter-tuning/. (last visited Mar. 28, 2025).

peanuts because they bought a Reese's Cup once. If the learning rate is too low, the model might take too long to adjust to patterns and fail to catch that someone who used to only buy peanut butter once a month now buys it weekly.

Hyperparameter tuning is a process of trial and error which seeks to find the ideal hyperparameter values to improve performance. Of course, one man's perfectly performing machine learning model is another man's misinformation spreading machine, so like all the other processes we have discussed, hyperparameter tuning is influenced by the overall goals of the system. As an IBM article explains, "Good hyperparameter tuning means a stronger performance overall from the machine learning model according to the metrics for its intended task."[120]

Unlike "regular" parameters, hyperparameters are not learned from the data. Instead, their values are set before training and determined through hyperparameter tuning.[121] A main goal of hyperparameter tuning is balancing bias and variance. Bias is the divergence between model predictions and reality, and it often occurs when models are under tuned, underfit, or otherwise unable to generalize well. Variance is the ability of a model to make consistent recommendations even when applied to non-training data, or its "sensitivity" to new data. Variance should be neither too low nor too high; if it is too high, a model struggles to generalize to new data and if it is too low, it is unable to capture underlying patterns in the training data resulting in overly simplistic predictions.

Model size and complexity make it unrealistic to manually "plug in" different values with success, particularly as hyperparameter landscapes can be noisy and unpredictable.[122] Even state-of-the-art models, such as large social media recommender systems, can perform poorly with arbitrarily chosen hyperparameter values, even if those hyperparameter values worked well on a similar model in the past. Furthermore, poor hyperparameter tuning can lead to misleading performance evaluations because they measure progress based on underperforming baselines.[123]

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[120] IBM, Ivan Belcic & Cole Stryker, *What Is Hyperparameter Tuning?*, IBM, https://www.ibm.com/think/topics/hyperparameter-tuning (last visited Mar. 28, 2025).
[121] Geeks for Geeks, *Hyperparameter Tuning*, GeeksforGeeks (2025), https://www.geeksforgeeks.org/hyperparameter-tuning/.
[122] B.G. Galuzzi et al., *Hyperparameter Optimization for Recommender Systems Through Bayesian Optimization*, 17 Computational Mgmt. Sci. 495 (2020), https://doi.org/10.1007/s10287-020-00376-3.
[123] Faisal Shehzad & Dietmar Jannach, *Everyone's a Winner! On Hyperparameter Tuning of Recommendation Models*, in *Proceedings of the 17th ACM Conf. on Recommender Systems (RecSys '23)* 652 (2023), https://doi.org/10.1145/3604915.3609488.

These issues have led to the development of hyperparameter tuning methods that employ computation to find optimal hyperparameters for large and complex models.

Learning rate, number of epochs, tree depth, and number of trees are just a few examples of hyperparameters used in machine learning models. Hyperparameters are chosen depending on model functionality and purpose; different types of models require different hyperparameters. Decision trees and random forests, two common prediction algorithms, use tree-related hyperparameters like tree depth and number of trees. On the other hand, a neural network does not use tree-specific hyperparameters like tree depth and number of trees. Instead, a neural network uses hyperparameters like number of layers and number of nodes per layer.

With hyperparameter choice established, the challenge becomes choosing optimal values: A model with a learning rate of 0.01 can perform quite differently than one with a learning rate of 0.1, because learning rate controls how much model weights are updated during training. To find the best value for a hyperparameter, engineers run many "experiments." In other words, they run the model over and over again with different values to find the best performance. Because there are multiple hyperparameters for each model, there are many possible combinations of hyperparameter values. Therefore, the hyperparameter tuning process can be computationally complex and techniques present trade-offs between effectiveness and expensiveness.

Common methods to find optimal hyperparameters include Grid Search, Random Search, and Bayesian optimization. The first two, Grid Search and Random Search, are given specified value ranges for each hyperparameter. While Grid Search systematically evaluates all combinations of hyperparameters within the predefined ranges, Random Search samples hyperparameters randomly from those ranges. Though Grid Search guarantees the best configuration within the search space, it is computationally expensive and inefficient in high-dimensional spaces. Meanwhile, though Random Search is more efficient in such spaces, it does not guarantee an optimal combination.

Bayesian optimization builds a probabilistic model to approximate the relationship between hyperparameters and performance, balancing exploration of new areas and exploitation of established areas using a mathematical function. It can be an effective method for recommender systems, such as in tuning collaborative-filtering matrix factorization hyperparameters.[124] Though they differ in methodology, the central feature of all hyperparameter tuning techniques is that they

[124] Galuzzi et al. *supra* note 122.

aim to improve performance by controlling the settings of a machine learning model. To restate a recurring point: exactly what "good performance" means depends on the metrics used to measure it.

Ranking Algorithms

A standalone algorithm does not necessarily optimize for engagement, but it does optimize for a given objective. If that objective happens to be engagement, the algorithm will optimize for engagement. The technical architecture of recommender systems for social media companies are proprietary, so we cannot know precisely which algorithms platforms use in their architecture. Furthermore, limited access to large-scale data challenges researchers' abilities to reverse engineer the systems.[125] We have already explored data processing, architecture, and hyperparameter tuning. However, each of these processes depends on "low-level" algorithms to complete their many tasks. Therefore, it is worthwhile to explore some of these algorithms to reasonably speculate about platforms' approaches. Through this analysis we may demonstrate how at every level, recommender systems are shaped by the overall goals of engineers and platforms that implement them.

We refer to the following algorithms as "low-level" because they are typically used as individual components within larger recommender system architecture, rather than functioning as full systems on their own. In contrast, "high-level" architecture like two-tower models integrate many algorithms into a broader framework. Therefore, understanding a recommender system requires a contextualization of low-level algorithms, even though they may be contained technical procedures.[126]

Whether past data is labeled or unlabeled determines whether the algorithm is considered supervised or unsupervised, a main classification of machine learning. For data to be labeled means that each point has been assigned a contextual tag or label that can be used for model training. In supervised machine learning, data is labeled before being fed into the model. In this way, the model learns by example. On the other hand, unsupervised machine learning algorithms perform

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[125] Paul Bouchaud, *Algorithmic Amplification of Politics and Engagement Maximization on Social Media*, in *Complex Networks & Their Applications XII* 145 (Hocine Cherifi et al. eds., vol. 1144, Springer 2024), https://doi.org/10.1007/978-3-031-53503-1_11.

[126] Bernhard Rieder et al., *From Ranking Algorithms to 'Ranking Cultures': Investigating the Modulation of Visibility in YouTube Search Results*, 24 Convergence 1 (2018), https://doi.org/10.1177/1354856517736982.

computational analysis to find patterns in unlabeled data. Social media platforms likely use both types of machine learning in their recommender systems for robustness and task-specification.

Supervised machine learning algorithms vary in complexity. Logistic regression is a simple example, used for binary classification. Because logistic regression is used to predict the probability of an event occurring, the dependent variable is bounded between 0 and 1. A logistic regression model applies a mathematical function called a sigmoid to a value that represents a combination of weighted and biased input features. By mapping this value into the bounded range, logistic regression will always output a probability. Logistic regression models learn, or improve performance, by adjusting the weights for each input feature and optimizing them to minimize the difference between predicted and actual outcomes. Recommender systems likely use logistic regression to predict binary types of user engagement, such as the presence of a like or click. By assigning weights to relevant features, the model can generate a likelihood score that informs the ranking algorithm about how to prioritize the content. The major issue with logistic regression is its assumptions: that the relationship between input and output is linear, that data is independent, and that there are no extreme outliers.[127] These assumptions limit the algorithm's ability to capture more complex engagement patterns such as positive versus negative engagement.

Another widely used supervised machine learning algorithm is decision trees. Decision trees offer a more flexible approach to classification tasks, because they partition data into hierarchical splits without any linearity assumption. They use a recursive, rule-based approach to split data at each node based on specific conditions, where branches from each node represent different categories or outcomes. For example, data could be split based on whether it has a like. Decision trees start at a root node and then partition the data according to whichever condition provides the greatest reduction in impurity. Impurity refers to the number of samples in a node that don't "fit" with the rest of the samples. Essentially, the goal of a tree is to end up with *n* leaves where data in each leaf belong to the same category, that is, each leaf has near-zero impurity. The tree continues branching until it reaches a stopping condition, which can be something like minimum number of samples per leaf or number of leaves *n*. The values of stopping conditions are often determined by hyperparameter tuning.

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[127] Rakesh Aggrawal & Priya Ranganathan, *Common Pitfalls in Statistical Analysis: Linear Regression Analysis*, 8 Persps. Clinical Rsch. 100 (2017), https://doi.org/10.4103/2229-3485.203040.

In recommender systems, decision trees may be used to rank content based on user engagement where each split represents an engagement-based signal. Once the model is trained (the tree is formed), the model assigns a score to each piece of content by following the branches of the tree. This traceability is an advantage of decision trees because it makes it easy to determine why a certain recommendation was made. However, decision trees are prone to overfitting, because they can keep splitting the data until every training example is neatly classified even if the splits themselves are based on irrelevant patterns.

Both logistic regression and decision trees rely on a single model for classification tasks, which can be problematic for generalizability and robustness. To improve overall predictive performance and stability, engineers often use ensemble methods, which combine multiple independent models.[128] For instance, Gradient Boosted Machines (GBMs) are a class of ensemble methods which combine many weak decision trees and successively improve the trees to build a stronger predictive model. GBMs train a series of decision trees sequentially, with each tree correcting the errors made by previous ones. To begin, a simple decision tree is fitted to the data. Predictions from this initial tree establish a baseline, and at each successive step the algorithm calculates the difference between the predicted and actual outcomes, called the residuals. The predictions from the initial tree may be quite poor, so a second tree is trained to predict the residuals of the first. Their predictions are then combined to generate an improved estimate. This process is repeated across many iterations, with each new tree learning to correct the mistakes of the previous ones. As long as more than half the decision trees are correct, the GBM will be correct.

GBMs are useful in ranking tasks because they can handle non-linear relationships and complex patterns better than simple models like logistic regression and vanilla decision trees. When engagement-based metrics like click through rate or watch time are chosen as target optimization objectives, each decision tree within the ensemble learns patterns that improve performance on such metrics. Thus, the entire model evolves to amplify content that optimizes its performance metrics. While GBMs are valuable in their interpretability and stability, they are often trained in batches. For short-form social media platforms, this poses a challenge for real-time suggestions.

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[128] Lior Rokach, Ensemble Methods for Classifiers, in *Data Mining and Knowledge Discovery Handbook* (Oded Maimon & Lior Rokach eds., Springer 2005), https://doi.org/10.1007/0-387-25465-X_45.

Different algorithms are well suited for different tasks: simple models like logistic regression are highly interpretable, enabling transparency and explainability, while gradient boosted machines are complex ensemble models that generally offer higher predictive accuracy. Many other relevant ranking algorithms, including DBSCAN, Hierarchical Clustering, and Spectral Clustering, are beyond the technical scope of this discussion. Regardless of model choice, ranking systems ultimately reflect platform priorities via their feature sets, optimization targets, and ranking algorithms. When engagement-based metrics serve as the primary optimization targets, each model is trained to detect and amplify content that maximizes those metrics. In the case of deep learning models, the same real-time responsiveness that allows them to personalize feeds also intensifies feedback loops. Though we cannot definitively say which algorithms or hyperparameters are used, the common thread between all discussed methods (and even those not discussed) is that their output is defined by the objectives of the engineers who implement them.

* * * *

Recommender systems are an amalgamation of algorithmic techniques, from collaborative filtering and content-based filtering to advanced architecture like two-tower, multimodal, and deep learning systems. Systems rely on different combinations and formats of input, whether user behavior, item features, or both, but they all produce the same kind of output: ranked suggestions personalized to individual users. These systems are further shaped through hyperparameter tuning and ranking algorithms such as decision trees, gradient boosted machines, and neural networks, which refine how content is prioritized at the lowest level.

Every stage of a recommender system—from feature engineering to model architecture to low-level algorithms—relies on some amount of human decision. Even the most complex and opaque systems reflect these choices, whether to maximize engagement, promote diversity, or align with civic design goals. These choices come predominantly from the platform, rather than the user, and remain largely unintelligible to the user. The effectiveness of a model is not judged by an objective standard of "good recommendations" but by how well it serves the platform's goals. In this way, recommender systems are not purely technical artifacts; they are socio-technical systems whose behavior can and should be shaped intentionally.

Chapter 5: The Legal Classification of Recommender Systems

While previous chapters discussed the technical and financial incentives behind engagement-driven algorithms, this chapter explores the legal landscape around recommender systems. Specifically, it will pose the question: Are algorithmic recommendations considered First Amendment protected speech, are they neutral tools, or are they both? The answer could reshape how platforms are regulated and how they design their recommender systems—and the makeup of user feeds.

The legal status of personalized recommender systems is currently ambiguous. In the United States, Section 230 of Title 47 of the United States Code (47 U.S.C. §230) grants online service providers (OSPs) broad immunity from liability for user-generated content. Courts have historically treated algorithmic curation as an extension of neutral functionality rather than editorial conduct,[129,130,131] allowing social media platforms to qualify for §230 protection. However, this logic has fallen under scrutiny as recommender systems become increasingly sophisticated and personalized.

In recent years, lawmakers have introduced bipartisan proposals to limit platform immunity and hold platforms accountable for algorithmically amplified harms,[132] but their efforts largely skirt the issue of sweeping §230 immunity. That democratically elected officials from both sides of the aisle recognize the importance of regulating these systems speaks to their relevance in American politics and culture. If algorithmic recommendations continue to be considered neutral tools, it would be prudent to reinterpret or amend §230 to account for the personalized, influential nature of recommender systems.

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[129] *Gonzalez v. Google LLC*, 598 U.S. 617 (2023).
[130] *Force v. Facebook, Inc.*, 934 F.3d 53 (2d Cir. 2019).
[131] *NetChoice, LLC v. Paxton* and *Moody v. NetChoice, LLC*, 603 U.S. 707 (2024).
[132] Eliminating Abusive and Rampant Neglect of Interactive Technologies (EARN IT) Act of 2023, H.R. 2732 & S. 1207, 118th Cong. (2023); Safeguarding Against Fraud, Exploitation, Threats, Extremism, and Consumer Harms (SAFE TECH) Act, H.R. 3421 & S. 299, 117th Cong. (2021); Online Freedom and Viewpoint Diversity Act, S. 4534, 116th Cong. (2020), https://www.congress.gov/bill/118th-congress/house-bill/2732; https://www.congress.gov/bill/117th-congress/house-bill/3421; https://www.congress.gov/bill/116th-congress/senate-bill/4534.

Origins and Intent of Section 230

The origins of §230 sprung from legal uncertainty around liability for user-generated content on online platforms in the early 1990s. The period saw a proliferation of new online service providers (OSPs) and internet service providers (ISPs). Users were pushing the limits of what they could—and could not—do online. Two court cases highlighted the challenges and inconsistencies in this arena: *Cubby, Inc. v. CompuServe Inc.*[133] and *Stratton Oakmont, Inc. v. Prodigy Services Co.*[134]

Cubby involved defamatory statements posted on a server hosted by CompuServe, an early online service provider (OSP). CompuServe operated and hosted various forums where users could post content, such as newsletters. Rumorville USA, one such newspaper, was available as part of the Journalism Forum. The plaintiffs in the case alleged that Rumorville USA contained false and defamatory remarks about their competing publication. The U.S. District Court for the Southern District of New York held that CompuServe was not liable for the defamatory statements since it had no editorial control over the content and therefore functioned as merely a distributor. Specifically:

> "CompuServe has no opportunity to review Rumorville's contents before DFA uploads it into CompuServe's computer banks, from which it is immediately available to approved CIS subscribers. CompuServe receives no part of any fees that DFA charges for access to Rumorville, nor does CompuServe compensate DFA for providing Rumorville to the Journalism Forum; the compensation CompuServe receives for making Rumorville available to its subscribers is the standard online time usage and membership fees charged to all CIS subscribers, regardless of the information services they use. CompuServe maintains that, before this action was filed, it had no notice of any complaints about the contents of the Rumorville publication or about DFA."

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[133] *Cubby, Inc. v. CompuServe, Inc.*, 776 F. Supp. 135 (S.D.N.Y. 1991).
[134] *Stratton Oakmont, Inc. v. Prodigy Servs. Co.*, 23 Media L. Rep. 1794 (N.Y. Sup. Ct. 1995).

The court reasoned that to impose liability on distributors who neither know nor had reason to know of the defamatory content would unduly burden the free flow of information.

The court came to the opposite conclusion in *Stratton Oakmont v. Prodigy*. Prodigy, another early OSP, hosted bulletin boards where users discussed financial topics. Unlike CompuServe, Prodigy explicitly advertised itself as family friendly, implemented content guidelines, and utilized human moderators and screening software to remove offensive material. An anonymous user posted statements alleging that Stratton Oakmont, a securities investment firm, had engaged in fraudulent activities. The New York Supreme Court (a trial court) concluded that Prodigy did qualify as a publisher due to its moderation efforts, thereby making itself liable for the user-generated content it hosted. The court distinguished Prodigy from CompuServe by emphasizing that unlike CompuServe, which performed no moderation, Prodigy's conscious choice to exercise editorial control amounted to an assumption of greater liability.

The conflicting outcomes of *Cubby* and *Stratton Oakmont* created uncertainty for OSPs around liability event though neither created a binding precedent.[135] Platforms were essentially "damned if they did, damned if they didn't" when it came to moderation. If they moderated content, they could face liability as publishers. If they did not moderate content, they risked the reputational consequences of hosting harmful or unlawful content.

In response, Representatives Christopher Cox (R, CA) and Ron Wyden (D, OR) introduced the Internet Freedom and Family Empowerment Act, which allowed providers to moderate content, in other words, exercise their First Amendment rights, without fear of increased liability.[136] The Act was incorporated as Section 509 of the Telecommunications Act of 1996 (Public Law 104-104), which amended the Communications Act of 1934 by adding Section 230 to the end of Part I of Title II. §230 provided OSPs immunity from civil liability for content posted by third-party users, allowing them to perform "good Samaritan" content moderation without worrying about publisher or speaker liability. The goal of §230 was to find a balance between enabling platforms to moderate harmful content and preventing the Internet form becoming the Wild West.

The language of 230(c)(1) is simple but highly consequential:

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[135] Federal district court and state trial court rulings lack precedential authority beyond the specific case at hand.
[136] Telecommunications Act of 1996, Pub. L. No. 104-104, § 509, 110 Stat. 56, 137 (1996).

> "No provider or user of an interactive computer service shall be treated as the publisher or speaker of any information provided by another information content provider."

In other words, platforms are shielded from liability for most user-generated content. The most popular sites today—Google, YouTube, Facebook, Wikipedia, Instagram, Reddit, Pornhub, TikTok, X, and Amazon—consist almost entirely of user-generated content.[137]

Without §230 protection, these platforms would be vulnerable to litigation for the defamation, harassment, terrorism, intimate image abuse, and other harms that they host. The pre-§230 cases of *Cubby* and *Stratton Oakmont* illustrate the dilemma: CompuServe avoided liability by not moderating at all, while Prodigy was held liable precisely because it attempted to moderate to provide a family-friendly experience—and failed. In this way, platforms risked being punished for trying to do the right thing. Section 230 aimed to alleviate this conundrum by provisioning that OSPs are free to moderate without the moderation itself leaving them liable to litigation over the content they host. It is hard to imagine an Internet, much less modern social media, without §230 because platforms would either be exceedingly restrictive in their moderation, or complete free-for-alls.

Early Interpretations of Section 230

Courts interpreted §230 immunity broadly in the years after enactment. The first major case to interpret §230 was *Zeran v. America Online, Inc.*[138] Kenneth Zeran sued America Online (AOL) after an anonymous user posted false and defamatory messages alleging his support for the Oklahoma City bombing six days prior. (Specifically, they were advertisements for T-shirts which celebrated the attack; the ads were attributed to Zeran.) The messages resulted in harassment and threats against Zeran; he argued that AOL was liable for failing to remove the messages in a timely manner. The Fourth Circuit Court of Appeals ruled that §230 protected AOL from liability, establishing a broad precedent for platform immunity:

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[137] Tiago Bianchi, *Global Top Websites by Unique Visitors 2024*, Statista (2025), https://www.statista.com/statistics/1201889/most-visited-websites-worldwide-unique-visits/.
[138] *Zeran v. Am. Online, Inc.*, 129 F.3d 327 (4th Cir. 1997).

> "By its plain language, § 230 creates a federal immunity to any cause of action that would make service providers liable for information originating with a third-party user of the service."

In its reasoning, the court held that imposing liability on AOL would create a chilling effect by overburdening service providers with lawsuits. The next case to be filed that required §230 interpretation ended in a ruling that closely tracked *Zeran*'s reasoning, despite even grimmer circumstances—a mother filed a complaint against AOL for hosting child sexual abuse material (CSAM) of her son in its chat rooms.[139,140] It was just the beginning. *Zeran* set the stage for courts to interpret §230 as providing near-comprehensive immunity to platforms, regardless of their moderation practices.

Another early case to interpret §230 was *Fair Housing Council of San Fernando Valley v. Roommates.com, LLC*[141] in the Ninth Circuit Court of Appeals. Roommate.com[142] operated a website designed to match individuals seeking roommates with one another. The site required users to disclose personal information, including gender, sexual orientation, and familial status and allowed users to specify roommate preference based on these qualities. The Fair Housing Council argued that such a practice violated federal and state housing discrimination laws.

The Ninth Circuit ruled that Roommate was not entitled to §230 protection for content it helped develop. According to the court, creating mandatory questionnaires and requiring users to provide personal information qualified Roommate as an information content provider because it rendered the site "responsible, in whole or in part, for the creation or development of" the offending content. Additionally, Roommate was not entitled to §230 immunity for "the operation of its search system, which filters listings." The court only granted immunity for the "Additional Comments" section in which users could type whatever they wanted with no guidance, clarifying a very important distinction for the OSPs: that there is a difference between simply providing an arena for user activity and actively directing or soliciting user activity within that arena. *Roomates.com* thus showed that by engaging in the latter—materially contributing to the development of unlawful

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[139] Patrick J. Carome & Cary A. Glynn, *Serendipity and Internet Law: How the 'Zeran v. AOL' Landmark Almost Wasn't*, The Recorder (Nov. 2017).
[140] *Doe v. America Online, Inc.*, 783 So.2d 1010 (F.L. Sup. Ct. 2001).
[141] *Fair Hous. Council of San Fernando Valley v. Roommates.com, LLC*, 521 F.3d 1157 (9th Cir. 2008) (en banc).
[142] A direct footnote from the opinion states: "For unknown reasons, the company goes by the singular name "Roommate.com, LLC" but pluralizes its website's URL, www.roommates.com."

content, such as structuring user interactions in ways that encourage illegal behavior—platforms risk losing §230 immunity for that part of their systems.

The line between providing an arena for users to post and actively directing their activity is not clearcut. The Ninth Circuit admitted in *Roommates.com* that "the broadest sense of the term "develop" could include the functions of an ordinary search engine—indeed, just about any function performed by a website." However, it says, such a broad reading of §230 would defeat the purpose of the legislation entirely. The court interpreted the term "development" as "referring not merely to augmenting the content generally, but to materially contributing to its alleged unlawfulness." But in many ways that leaves us in the same place, because what constitutes "material contribution to development" was not, and has not been, entirely resolved.

The question of how to apply the material contribution test remains open. Even sites that actively organize or filter illegal content can receive §230 immunity; the level or prominence of illegal content may have no bearing on a site's liability. A particularly notable instance of this protection is *Jane Doe No. 1 v. Backpage.com, LLC.*[143] The plaintiffs in the case, survivors of human trafficking, argued that Backpage.com had "deliberately structured its website to facilitate sex trafficking." The First Circuit ruled that §230 shielded Backpage.com from liability, affirming that because the platform did not create and structure the content itself—such as in *Roommates.com*—in this case, immunity applied. The circuit judge admitted that the case was difficult because the law required "that we [..] deny relief to plaintiffs whose circumstances evoke outrage."

Though the aforementioned cases involve §230, none of them concern personalized recommender systems. The courts' views of OSPs are confined to websites that have users log on and expressly interact with other users, unlike the infinite scroll feeds of today which function as products in and of themselves. Accordingly, the rise of algorithmic curation has sparked intense legal and political debate over regulation and accountability. A recurring allegation is that a platform goes beyond passively hosting content when it actively targets content to users.[144,145] Recent cases pit the traditional neutral conduit view of platforms against claims that the output of recommender systems amounts to expressive First Party speech, with a few judges arguing that by

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[143] *Jane Doe No. 1 v. Backpage.com, LLC*, 817 F.3d 12 (1st Cir. 2016).
[144] *Anderson v. TikTok, Inc.*, 22-3061 F.4d ___ (2024).
[145] *Twitter, Inc. v. Taamneh*, 598 U.S. 471 (2023).

engaging in First Amendment-protected conduct, platforms lose their §230 immunity. However, this argument remains very much outside of the conventional jurisprudence and in general courts remain uncertain as to where to draw the line in terms of platform protections.

We will take a brief but necessary detour here to discuss the concept of *common carriage.* Common carriage traditionally applies to industries that provide essential services to the public, such as railroads, telecommunications providers, and utilities. The origins of common carriage extend as far back as Plantagenet England; modern versions require entities that serve the public to do so indiscriminately so long as customers comply with basic terms of service.[146] It is important to address in the context of social media because of its ubiquity; the widespread adoption of a few platforms has made it arguably essential to modern life.

However, commercial online services do not clearly meet the strict legal definition of a common carrier, or "a person or a commercial enterprise that transports passengers or goods for a fee and establishes that their service is open to the general public."[147,148] The debate over whether social media platforms should be classified as common carriers has been central to discussions about platform regulation. Legal scholars offer competing interpretations. Professor Christopher S. Yoo argues that social media companies do not meet the legal criteria for common carriage because they exercise editorial discretion in curating and moderating content. Specifically, he dismisses four commonly cited justifications for doing so—public interest, monopoly power, inclusion in the communications industry, and government benefits—and argues that the only valid criterion is whether a company holds itself out as serving all members of the public indiscriminately. Since social media platforms enforce Terms of Service and selectively moderate content, they fail this test and cannot be regulated as common carriers.[149,150]

In contrast, Professors Morgan Ricks and Ganesh Sitaraman assert that common carriage is a flexible legal concept that has evolved over centuries to accommodate new technologies. They argue that businesses providing essential services to commerce and tending toward monopoly—

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[146] Steven W. Kopp & Mike Landry, *The Tangled Roots of Regulation in Marketing: A History of Common Calling*, in *Marketing History Knows No Boundaries* 8 (Carleton Univ. 1997).
[147] Cornell L. Sch., *Common Carrier*, Legal Info. Inst. (2021), https://www.law.cornell.edu/wex/common_carrier.
[148] Matthew C. Siderits, *Defamation in Cyberspace: Reconciling Cubby, Inc. v. CompuServe, Inc. and Stratton Oakmont v. Prodigy Services Co.*, 79 Marq. L. Rev. 1065 (1996).
[149] Christopher S. Yoo, *What's in a Name?: Common Carriage, Social Media, and the First Amendment*, 119 Nw. U. L. Rev. 49 (2024).
[150] Christopher S. Yoo, *Brief of Professor Christopher S. Yoo as Amicus Curiae in Support of Respondents in No. 22-277 and Petitioners in No. 22-555* (2023).

such as wharves, railroads, and telegraph companies—have historically been subject to common carriage obligations. Ricks and Sitaraman contend that social media platforms, like historical common carriers, have become essential to modern communication and economic life, thereby opening them up to common carrier regulation.[151] Their perspective fundamentally differs from Yoo in how they interpret the concept of common carriage in relation to the First Amendment. While Yoo emphasizes the role of editorial control as a disqualifying quality, pointing to Supreme Court precedents such as *Miami Herald v. Tornillo*, Ricks and Sitaraman argue that common carriage regulation has always allowed reasonable exclusions. In their view, social media can be such an exclusion.

Ultimately, courts have sided with Yoo's argument, ruling that social media platforms are not common carriers. For instance, in *NetChoice, LLC v. Attorney General, State of Florida* (2022), the Eleventh Circuit found that social media platforms "have never acted like common carriers" because of their ability to condition user access through Terms of Service.[152] While Ricks and Sitaraman present a compelling historical analogy, their argument has not gained traction in courts, which continue to view editorial control as the key factor distinguishing social media from traditional common carriers.

Algorithmic Curation under Section 230

A central question in §230 liability is whether algorithmically curated feeds should be considered distinct products—separate from the user-generated content they display. When they are viewed as distinct from third-party content, a small minority of judges have found that algorithmic curation constitutes first-party content, thus making it ineligible for §230 protection (since the legislation only shields platforms for third-party content).[153,154] However, the majority of jurisprudence holds platform moderation as protected under §230 when such moderation is considered part of the platform's neutral role in displaying user-generated content. These differing

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[151] Morgan Ricks & Ganesh Sitaraman, *What Centuries of Common Law Can Teach Us About Regulating Social Media*, L. & Pol. Econ. Project (2024), https://lpeproject.org/blog/what-centuries-of-common-law-can-teach-us-about-regulating-social-media/.
[152] *NetChoice, LLC v. Att'y Gen., State of Fla.*, 34 F.4th 1196 (11th Cir. 2022).
[153] *Anderson supra* note 144.
[154] *Force v. Facebook, Inc.*, 934 F.3d 53 (2d Cir. 2019), (Katzmann, C.J., dissenting).

opinions demonstrate how the growing complexity of personalized recommender systems leads to uncertainty both between and within the judicial system.

As in *Fair Housing Council of San Fernando Valley*,[155] if a platform is deemed to have "materially contributed" to the unlawful content, they are not eligible for §230 protection. However, in the case of recommender systems, the definition "material contribution" has not been so clear. In *Force v. Facebook,*[156] families of the victims of a terrorist attack sued Facebook under the Antiterrorism Act of 1992 (ATA). The plaintiffs claimed that Facebook's algorithms "connect[ed] terrorist sympathizers with pinpoint precision" and allowed extremist organizations to strategically recruit based on demonstrated interest. In other words, Facebook had materially contributed to the attack of their family members by designing its feed in such a way that facilitated terrorism. The Second Circuit majority nevertheless treated these claims as fundamentally targeting Facebook's role as a publisher of third-party content. The Court even described how Facebook's personalized "News Feed" algorithm worked at the time: "The newsfeed algorithms—developed by programmers employed by Facebook—automatically analyze Facebook users' prior behavior on the Facebook website to predict and display the content that is most likely to interest and engage those particular users." Crucially, the court found that the functions of the algorithms—tailoring and sorting user content—constituted "making information more available" and thus protected: "We find no basis in [§230] for concluding that an interactive computer service is not the "publisher" of third-party information when it uses tools such as algorithms that are designed to match that information with a consumer's interests." *Force* illustrates a key conundrum in the "material contribution" test, in that the line between contributing to unlawful content and enabling the free flow of information is not always so clear.

Similarly, in *Dyroff v. Ultimate Software Group,*[157] the Ninth Circuit dismissed a lawsuit against a platform whose recommender system directed a user to an illicit drug forum, ultimately leading to their fatal overdose. The court applied the three-pronged Barnes Test, which asks "(1) whether a defendant is a provider or user of an interactive computer service (2) whom a plaintiff seeks to treat, under a state law cause of action, as a publisher or speaker (3) of information provided by another information content provider."[158] Because the answer to all three prongs was

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[155] *Fair Hous. supra* note 141.
[156] *Force v. Facebook, Inc.*, 934 F.3d 53 (2d Cir. 2019).
[157] *Dyroff v. Ultimate Software Grp.*, 934 F.3d 1093 (9th Cir. 2019).
[158] *Barnes v. Yahoo!, Inc.*, 565 F.3d 560 (9th Cir. 2009).

affirmative, the court held that Ultimate Software Group was entitled to §230 protection. In both *Dyroff* and *Force*, the courts affirmed broad immunity for algorithmic ranking of user posts because such ranking was deemed a content-neutral function, regardless of the complexity and personalized nature of the algorithms.

Complexity and personalization are important because they theoretically inform the distinction between simply arranging others' content versus developing it. In *Roommates.com* the court denied §230 immunity on the grounds that Roommates.com required users to answer predefined questions about roommate preferences and then used those answers to filter search results. This rudimentary form of personalization required active user input; because the site itself elicited and used unlawful criteria under the state and federal fair housing laws, it became responsible for the discriminatory content users posted in response.[159] In contrast, as *Force, Backpage.com,* and *Ultimate Software Group* demonstrate, complex, personalized algorithms which do not require active input from users besides their engagement may be considered simply "neutral tools" that do not amount to development.

The "material contribution" test has been widely adopted,[160] but modern recommender systems seem to fall outside the binary of 'either this one or that one.' Though these systems do not explicitly guide user action, they do collect vast volumes of user data to inform their recommendations. Platforms are largely unusable without sustained user activity, because user engagement helps train the recommender systems that make the platforms so appealing. In practice, courts have been reluctant to situate algorithmic curation over the "development" line. However, when considering the methodology and objectives of personalized recommender systems, the binary of "material contributor/publisher" or "neutral tool" is fundamentally outdated and fails to reflect the realities of modern algorithmic design.

Editorial Control and First Amendment Protection

In general, classification becomes more difficult without a binary. This is particularly true for legal and political experts attempting to apply established case law to evolving technical processes. Therefore, we will return to the binary and to explore the editorial control side of social

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[159] *Fair Hous. supra* note 141.

[160] Megan Iorio & Tom McBrien, *EPIC Traces the Ninth Circuit's Surprisingly Narrow Section 230 Interpretation as the Court Hears Two Important Cases*, Elec. Priv. Info. Ctr. (Apr. 2024), https://epic.org/epic-traces-the-ninth-circuits-surprisingly-narrow-section-230-interpretation-as-the-court-hears-two-important-cases/.

media platforms. As discussed, platforms receive broad §230 immunity when their algorithms are considered neutral infrastructure. But what if they are not considered neutral infrastructure, and rather an extension of editorial discretion protected under the First Amendment? By considering the origins of editorial control and §230, we see that two things can be true at once: a platform can exercise First Amendment-protected editorial judgment over content while retaining §230 protection.

The roots of First Amendment protection for editorial control long predate §230. Two concepts are particularly significant in establishing constitutional protection for editors and publishers: 1) That editorial control is protected under the First Amendment and 2) that editors and publishers are protected from liability for factual inaccuracies about public figures unless they act with actual malice, meaning they knowingly publish false information or show reckless disregard for the truth. For private individuals, however, a lower standard applies. Liability may arise if the statement is false, defamatory, and presented as fact—regardless of the speaker's intent.[161]

The Supreme Court has repeatedly reinforced the principle of editorial independence as a core component of press freedom. In *New York Times Co. v. Sullivan,*[162] a landmark case involving a defamation suit brought by a Montgomery, Alabama police commissioner. The New York Times had published a full-page advertisement describing civil rights protests in Montgomery, praising Dr. Martin Luther King, and naming various Southern officials who allegedly engaged in civil rights violations. Several factual errors in the advertisement became the basis for a defamation lawsuit by the police commissioner. The Court ruled unanimously for the Times; by establishing a high bar for defamation claims by public officials, the ruling afforded editors and publishers greater freedom to cover controversial issues without constant fear of lawsuits.

The principle was further strengthened in *Miami Herald v. Tornillo,* in which the Court struck down a Florida law that required newspapers to publish a political candidate's reply to critical editorials. The Court held that newspapers cannot be forced to provide a "right of reply," ruling against the candidate. In other words, the government cannot force an editor or publisher to print content. By doing so, it emphasized editorial control as an essential component of press freedom.

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[161] *Gertz v. Robert Welch, Inc,* 418 US 323 (1974).
[162] *New York Times Co. v. Sullivan,* 376 US 254 (1964).

Compelled speech—which refers to the protection of an individual from being required (compelled) to utter or otherwise express a thought with which that individual disagrees—is another issue important to discussing social media responsibility and regulation. The foundational case in this doctrine is *West Virginia State Board of Education v. Barnette,*[163] in which the Supreme Court held that public schoolchildren could not be forced to salute the flag or recite the Pledge of Allegiance. The Court emphasized that the First Amendment protects both the right to speak and the right *not* to speak—to not be compelled—to affirm beliefs one does not hold.

Later cases built on this principle in the context of media and corporate speech. In *Pacific Gas & Electric Co. v. Public Utilities Commission*,[164] the Court ruled that a private utilities company could not be required to include third-party speech with which it disagreed in its billing envelopes. The justices grounded their decision in what they called affirmative First Amendment rights: "For corporations, as for individuals, the choice to speak includes within it the choice of what not to say." Whether prioritizing engagement through recommender systems is an affirmative editorial choice has been contested.[165,166]

The *Barnette* doctrine was challenged by the evolving technologies of television. *Red Lion Broadcasting Co. v. FCC*[167] introduced the idea that editorial control may be subject to regulation when a medium serves as a scare or monopolistic gateway to public discourse. In *Red Lion*, the Court upheld the FCC's Fairness Doctrine, which required broadcasters to give each side of public issues fair coverage when presenting such issues. It reasoned that the limited nature of the broadcast spectrum justified requiring broadcasters to air opposing viewpoints. *Red Lion* acknowledged that certain media possess structural power that can suppress diverse speech, thus the Court prioritized the public's right to access a range of information over the platform's unrestricted editorial discretion.

In *Turner Broadcasting v. FCC*,[168] the Court ruled that cable television operators may be subject to must-carry regulations because they act as gatekeepers able to restrict access to

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[163] *West Virginia State Board of Education v. Barnette*, 319 US 624 (1943).
[164] *Pac. Gas & Elec. Co. v. Pub. Utils. Comm'n*, 475 U.S. 1 (1986).
[165] Robert Diab, *The Case for Mandating Finer-Grained Control Over Social Media Algorithms*, Tech Pol'y Press (2024), https://www.techpolicy.press/the-case-for-mandating-finergrained-control-over-social-media-algorithms/.
[166] Grafanaki, Sofia. 2018. "Platforms, the First Amendment and Online Speech: Regulating the Filters." Pace Law Review 39, no. 1 (September): 1-52.
[167] *Red Lion Broadcasting Co. v. FCC,* 395 US 367 (1969).
[168] *Turner Broad. Sys., Inc. v. FCC*, 512 U.S. 622 (1994).

competing content. The Court cites *Red Lion*: "As we said in *Red Lion,* "[w]here there are substantially more individuals who want to broadcast than there are frequencies to allocate, it is idle to posit an unbridgeable First Amendment right to broadcast comparable to the right of every individual to speak, write, or publish." The Court determined that the cable operators' bottleneck amounted to complete control over what was and was not shown. Furthermore, the must-carry provisions of the legislation did not force cable operators to change their *own* messages to respond to broadcast programming, unlike the requirements imposed by the Public Utilities Commission in the *Pacific Gas & Electric* ruling. Thus, the legislation did not limit editorial control. The Court also distinguished its ruling from *Miami Herald*, because unlike newspaper editors, cable operators can silence the voice of competing speakers "with the mere flick of a switch."[169]

A fundamental tension in the legal treatment of social media platforms is highlighted by these cases. Looked at in aggregate, the cases support platforms' claims that through affirmative First Amendment rights, editorial discretion protects them from being forced to host or promote certain speech by the government, as well as from wanton civil action. In other contexts, platforms contend that they are protected under Section 230 from civil liability expressly because their recommender systems are simply neutral infrastructure on which they host content. In other words, Section 230 applies when platforms are viewed as neutral tools, and the First Amendment applies when they are viewed as editors. However, one might ask, if they are neutral tools, what happens when the content they predominantly promote is unlawful? And as recommender systems become more complex, influential, and automated, can the design and operation of these systems themselves be additionally considered a form of expressive speech entitled to First Amendment protection?

To qualify for First Amendment protection, speech must convey an idea—in other words, it must be more than purely functional.[170,171] Editorial control is protected because editors are expressing themselves through their editorial discretion. Something that lacks an expressive element (is purely functional) is not eligible for First Amendment protection. Thus, the question:

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[169] *Turner supra* note 168.
[170] *Gibney v. Empire Storage & Ice Co.,* 336 U.S. 490 (1949).
[171] *Spence v. Washington*, 418 U.S. 405 (1974).

What makes something expressive instead of purely functional? Well, even the Founders could not explicitly agree upon the concept of expressive freedom.[172]

The first Supreme Court Case to expressly deal with the question of protected speech was over a hundred years ago. In *Schenck v. United States*, Justice Holmes introduced the famous "yelling fire in a crowded theater" aphorism.[173] Later, in *Roth v. United States* the Court explained that "all ideas having even the slightest redeeming social importance," including those concerning "the advancement of truth, science, morality, and arts" are fully protected by the First Amendment.[174] Now in the digital age, courts have grappled with this question in the context of software and algorithms. In this realm, the line between expressive and functional is often quite blurry.

The Second Circuit held that code used to circumvent encryption was both functional and expressive in *Universal City Studios v. Corley*.[175] The code was functional because it performed a mechanical task, and it was expressive because writing and distributing code inherently involves communicating ideas and technical information. The ruling relied on reasoning from *Bernstein v. U.S. Department of Justice*.[176] In that case, the Ninth Circuit struck down a piece of regulation due to its risk of "interdicting the flow of scientific ideas (whether expressed in source code or otherwise.)" The court found that source code for encryption software was protected under the First Amendment because it reflected the author's ideas on cryptography. However, it emphasized the narrowness of its First Amendment holding: "We do not hold that all software is expressive. Much of it surely is not." Similarly, in *Junger v. Daley,* the Sixth Circuit explained that "Because computer source code is an expressive means for the exchange of information and ideas about computer programming, we hold that it is protected by the First Amendment."[177]

That the Sixth Circuit based its protection of computer source code on the classification that such code is expressive speech reflects a system of judicial review that courts use to assess whether a law that affects constitutional rights is permissible. This system is called judicial

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[172] Jud Campbell, *Natural Rights and the First Amendment*, 127 Yale L.J. 246 (2017), https://www.yalelawjournal.org/article/natural-rights-and-the-first-amendment.
[173] *Schenck v. United States*, 249 U.S. 47 (1919) (effectively overruled by *Brandenburg v. Ohio,* 395 U.S. 444 (1969)).
[174] *Roth v. United States*, 354 U.S. 476, 484 (1957).
[175] *Universal City Studios, Inc. v. Corley*, 273 F.3d 429 (2d Cir. 2001).
[176] *Bernstein v. U.S. Dep't of Just.*, 945 F. Supp. 1279 (N.D. Cal. 1996), *but* holding was effectively mooted by subsequent regulatory changes that relaxed cryptography export control rules.
[177] *Junger v. Daley*, 209 F.3d 481 (6th Cir. 2000).

scrutiny, and courts use it to determine whether the government's interest in enacting regulation is sufficient to justify the law's intrusion on individual rights. In the context of the First Amendment, courts typically apply one of three levels of scrutiny: rational basis, intermediate scrutiny, or strict scrutiny. In its reasoning in *Junger,* the Court applied strict scrutiny, but courts do not always apply such a rigorous standard. The level of scrutiny applied depends on the nature of the regulation and the type of speech involved.

Under rational basis review, the government only needs to show that there is a rational connection between the law and a legitimate state interest. This standard is highly deferential to the government and is generally used when neither a fundamental right nor a suspect classification is at stake. When the rational basis test is used, it rarely invalidates legislation.[178]

Intermediate scrutiny applies to regulations that affect speech in a more limited or indirect way, such as content-neutral time, place, and manner restrictions like noise ordinances. When expressive and non-expressive elements are combined—such as in the use of source code or algorithmic systems that both communicate ideas and produce functional output—courts may apply intermediate scrutiny. To satisfy intermediate scrutiny, the government must show that the law serves an important or substantial government interest, is unrelated to the suppression of expression, and is narrowly tailored so that the incidental restriction on speech is no greater than necessary to further the government interest.[179] Crucially, courts must first conclude that intermediate scrutiny is the appropriate standard, which is when the regulation in question targets the conduct or function rather than the expression itself.

The highest standard of constitutional review, known as strict scrutiny, applies to laws that restrict speech based on content or viewpoint, or that burden core expressive conduct protected by the First Amendment. To survive strict scrutiny, the government must prove that the regulation is narrowly tailored to serve a compelling government interest and that it is the least restrictive means of furthering that interest. Courts are especially cautious when expressive conduct is a stake, such as political protest. In such cases, the government is relatively unlikely to succeed in enacting the law.[180]

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[178] Raphael Holoszyc-Pimentel, *Reconciling Rational-Basis Review: When Does Rational Basis Bite?*, 90 New York Univ. L. Rev. 6 (December 2015).
[179] *United States v. O'Brien,* 391 U.S. 367 (1968).
[180] Thomas A. Berry, *The Supreme Court Grapples with Free Speech Scrutiny*, Cato Institute (January 16, 2025), https://www.cato.org/blog/supreme-court-grapples-free-speech-scrutiny.

An important part of the judicial review process for assessing a law's constitutional burdens is the nature of the speech. *United States v. Bernstein, Junger v. Daley,* and *Universal City Studios v. Corley* establish that while it may be functional, code can also be expressive and therefore qualify for First Amendment protection—and it can even be functional *and* expressive. This raises an important question in the case of algorithmic recommender systems: If writing and distributing code is protected as expressive activity, could designing and implementing algorithms as machine learning models that rank and promote content also be considered expressive?

Recent rulings have avoided answering this question outright, even when circumstances invite it. For instance, two companion suits from 2022, *Twitter, Inc v. Taamneh* and *Gonzalez v. Google LLC,* involved the role of the recommender systems in propagating harmful content. The cases focused on separate terrorist attacks in Europe. Plaintiffs alleged that Google and Twitter had not only knowingly allowed ISIS to use their recommender systems to recruit, fundraise, and propagandize, but also the platforms profited from advertisement placed on ISIS content. The Court concluded in *Twitter* (through which *Gonzalez* was resolved) that neither company was liable for the amplification of terrorist content, not because of §230 but because it did not knowingly and substantially assist terrorists. Thus, the Court concluded this did not qualify as illegal conduct under the Antiterrorism Act (ATA). In other words, because the platform did not encourage or coordinate with ISIS, the mere presence and amplification of terrorist content was not enough to establish liability under the ATA.

Although the Court resolved *Gonzalez v. Google* and *Twitter v. Taamneh* without directly addressing §230, the opinions contain many references which provide an understanding of how the Court views recommender systems. For example, the *Twitter* opinion acknowledges that recommendation algorithms "match any content with any user who is more likely to view that content, and that the platforms perform little to no front-end screening on any content before it is uploaded."[181] It further states that recommendation algorithms are "merely part of the infrastructure through which all the content on their platforms is filtered."[182] In summary, the Court concluded that despite the amplification of ISIS content, the platforms were not liable because they did not provide the terrorists with special treatment or explicit encouragement—in other words, their recommendation systems did not materially contribute to the unlawful conduct. While

[181] *Twitter, Inc. v. Taamneh*, 598 U.S. 471 (2023).
[182] *Id.*

certainly a reasonable lens through which to examine recommender systems on social media, the rulings were specific to the details of the case—the ATA and not §230—thus narrowing the scope of its applicability.

Outside of the material contribution framework, *Moody v. NetChoice* and *Paxton v. NetChoice*, *LLC* affirmed that platforms engage in expressive activity when curating content through moderation and ranking. Though it remanded both cases for further facial analysis, the Court offered insight into its view that platforms receive First Amendment protection when they compile and curate third-party speech into an expressive product. In the majority opinion for *Moody*, Justice Elena Kagan wrote: "When the platforms use their Standards and Guidelines to decide which third-party content those feeds display, or how the display will be ordered and organized, they are making expressive choices. And because that is true, they receive First Amendment protection." [183] That this quote directly contradicts the earlier one from *Twitter* is evidence of the Court's struggle to resolve a fundamental tension: If platforms exercise editorial control over what users see, is it possible to hold them responsible for the consequences of their curation—such as the systematic amplification of misinformation and extremism?

Evolving Interpretations

A central question in the §230 debate over recommender systems is whether platforms are simply exercising a publisher's traditional editorial function or effectively creating new content through algorithmic curation of user feeds. §230 immunity is lost if the platform is responsible for the creation of the unlawful content, but not if the platform is simply exercising editorial control. As we have discussed, neutral conduits are analogous to a typesetter arranging letters to the editor or a librarian organizing books. In *Force v. Facebook, Inc.*, the Second Circuit cited this analogy explicitly, stating that "arranging and distributing third-party information inherently forms "connections" and "matches" among speakers, content, and viewers of content, whether in interactive internet forums or in more traditional media."[184] On the other hand, *Roommates.com* affirmed that if a platform materially contributes to the illegality of the content, it becomes a developer of that content.

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[183] *Moody v. NetChoice, LLC*, 603 U.S. 707 (2024).
[184] *Force supra* note 156.

Lemmon v. Snap, Inc.[185] demonstrates a new iteration of the "material contribution" paradigm. The case involved the surviving parents of two boys who died in a high-speed car accident; the parents alleged that Snapchat, through "the interplay between Snapchat's reward system and its Speed Filter," incentivized reckless driving that killed their sons. The Ninth Circuit denied §230 immunity because the claim was based on Snap's own product design (the filter) rather than content from another user. Snap was being sued for creating a dangerous feature, not for publishing third-party content. Thus *Lemmon* illustrates the "material contribution" limit: when the design or functionality itself is at issue, §230 offers no protection, analogous to the blank demographic question boxes in *Fair Housing Council of San Fernando Valley*.[186] In the context of algorithmic personalization, this case suggests that if an algorithm is deemed a "defective product," a plaintiff might bypass §230 (though First Amendment protection could still apply).

In a partial dissent in *Force*, Judge Robert Katzmann argued that Facebook's friend and content recommendation algorithms went beyond publishing because they amounted to an affirmative role in facilitating harm. He analogized the algorithms to a person who studies two authors' works, then persistently suggests to one of them that they meet the other based on both of their work. Katzmann argues that the acquaintance might think of himself as a matchmaker—but he is not a publisher of either author's works: "It strains the English language to say that in targeting and recommending these writings to users—and thereby forging connections, developing new social networks—Facebook is acting as "the publisher of…information provided by another information content provider." 47 U.S.C. §230(c)(1)."[187] He concluded that "When a plaintiff brings a claim that is based not on the content of the information shown but rather on the connections Facebook's algorithms make between individuals, the CDA does not and should not bar relief."[188]

While Judge Katzmann authored his dissent from the parts of the *Force* opinion that compare recommender systems to publishers, the majority ruling granted Facebook §230 immunity. In mid-2024, the Third Circuit used similar logic in its ruling in *Anderson v. TikTok*.[189] Tawainna Anderson sued TikTok after her ten-year-old daughter tragically died while attempting

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[185] *Lemmon v. Snap, Inc.*, 995 F.3d 1085 (9th Cir. 2021).
[186] *Fair Hous. Council of San Fernando Valley v. Roommates.com, LLC*, 521 F.3d 1157 (9th Cir. 2008) (en banc).
[187] *Id.*
[188] *Id.*
[189] *Anderson v. TikTok, Inc.*, 22-3061 F.4d ___ (2024).

the viral "Blackout Challenge." The lawsuit alleged TikTok's algorithm "promoted [dangerous challenge] videos to minors" and knew such content was causing harm. A district court, treating the video as third-party content, had dismissed the case under §230. The Third Circuit reversed, holding that TikTok's recommendation algorithm was TikTok's own content (or conduct), not that of another information provider: "TikTok makes choices about the content recommended and promoted to specific users, and by doing so, is engaged in its own first-party speech." According to the Third Circuit, §230 immunity only covers third-party speech; if the platform's algorithmic choices are deemed first-party conduct, immunity does not apply. In other words, personalized recommendations are distinct from neutral hosting.

In reaction to the ruling, TikTok argued that this reasoning improperly characterizes standard feed algorithms as the platform's own speech, warning that it undermines §230 core protections. Several prominent online free speech organizations have voiced their support for TikTok in an amicus brief.[190] Nonetheless, the Third Circuit dismissed TikTok's appeal. Legal scholars observe that *Anderson* conflicts with past jurisprudence, particularly around their designation of platforms as publishers or speakers.[191,192] They argue that the purpose of the legislation was to ensure that services like Prodigy, which exercised editorial control over their platforms, were protected *even when* they hosted unlawful content. In contrast, the court in *Anderson* seems to suggest that platforms can *lose* their §230 when they exercise editorial control. It remains to be seen whether the Supreme Court will review the split between the Third Circuit and the First, Second, and Ninth Circuits.[193] This case highlights judicial confusion about applying blanket immunity or liability to algorithmic amplification of illegal behavior.

These dueling perspectives show us that the same technology can be viewed very differently. The disparities are exacerbated by the opacity and secrecy around actual technical implementation of the systems. Courts describe the technical functions of recommendation

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[190] Kate Ruane, Samir Jain & Becca Branum, *CDT Joins Amicus Brief Urging Rehearing in Anderson v. TikTok*, Ctr. for Democracy & Tech. (2024), https://cdt.org/insights/cdt-joins-amicus-brief-urging-rehearing-in-anderson-v-tiktok/.

[191] Tom McBrien, *In Anderson v. TikTok, the Third Circuit Applies Questionable First Amendment Reasoning to Arrive at the Correct Section 230 Outcome*, Elec. Privacy Info. Ctr. (2024), https://epic.org/in-anderson-v-tiktok-the-third-circuit-applies-questionable-first-amendment-reasoning-to-arrive-at-the-correct-section-230-outcome/.

[192] Ryan Calo, Jackie O & Kenneth S. Abraham, *Courts Should Hold Social Media Accountable — But Not by Ignoring Federal Law*, Harv. L. Rev. Blog (2024), https://harvardlawreview.org/blog/2024/09/courts-should-hold-social-media-accountable-but-not-by-ignoring-federal-law/.

[193] Cong. Rsch. Serv., *The United States Courts of Appeals: Background and Circuit Splits from 2024* (n.d.).

algorithms in varying detail, with some opinions delving deeper than others into how content is filtered and suggested to assess whether those processes are "neutral" or not. Katzmann argues that some recommendations result in the user becoming part of a "unique global community, the creation and maintenance of which goes far beyond and differs in kind from traditional editorial functions."[194] In *Anderson*, the Third Circuit focuses on TikTok's infamous "For You Page," or FYP: "TikTok users can search the platform for content or, without searching, view content that TikTok's algorithm recommends by posting the content to a user's [FYP]. TikTok's algorithm is not based solely on a user's online inputs. Rather, the algorithm curates and recommends a tailored compilation of videos for a user's FYP based on a variety of factors, including the user's age and other demographics, online interactions, and other metadata." It explained in a footnote: "We reach [our] conclusion specifically because TikTok's promotion of [Blackout Challenge videos] was not contingent upon any specific user input. Had [the victim] viewed a Blackout Challenge video through TikTok's search function, rather than through her FYP [feed], then TikTok may be viewed more like a repository of third-party content than an affirmative promoter of such content."[195] The court here makes a key distinction between active and passive user behavior. The court saw something qualitatively affirmative in TikTok's process. It sees the recommender system not as a neutral tool to organize user input, but as an engine driving content to users in a way that a passive bulletin board would not.[196]

The Third Circuit and Katzmann recognized that modern recommendation systems are sophisticated, data-driven engines that personalize content for each user.[197] The prevailing view in other jurisprudence is that these algorithms are fundamentally doing what editors have always done—deciding "to whom" and "in what order" content is shown, thus remaining protected under the First Amendment. Regardless, the legal landscape around recommender systems shows us that when a court focuses on the modern implementation and output of the system, the line blurs

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[194] *Force supra* note 156.
[195] *Anderson supra* note 189.
[196] Nowadays, news publishers take similar action when using algorithms to arrange their headlines on the front page.
[197] In his Force dissent, Katzmann describes Facebook's algorithms as "sophisticated" and explains: "After collecting mountains of data about each user's activity on and off its platform, Facebook unleashes its algorithms to generate friend, group, and event suggestions based on what it perceives to be the user's interests. [..] Facebook uses the algorithms to create and communicate its own message: that it thinks you, the reader—you, specifically—will like this content. [..] Facebook's suggestions contribute to the creation of real-world social networks."

between neutral tool and publisher/material contributor, opening the door for new interpretations and conflicting perspectives.

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It is important to note that §230 was enacted in 1996, in the era before ultra-personalized feeds. The earlier platforms at issue in *Cubby* and *Stratton Oakmont* were primarily composed of message boards and forums. They were far from the engagement-driven recommender systems that define modern social media. Facebook was not founded until 2004, TikTok until 2016, and Instagram Reels and YouTube Shorts until 2020. §230's authors could not have anticipated the degree to which personalized recommender systems would shape public discourse.

Personalized recommender systems have pushed the boundaries of how we think about intermediary liability for third-party content. They have forced courts to interpret §230 in scenarios unimaginable in 1996 by blurring the line between passive distribution of content and active editorial control. (Had §230 been passed just a few years later, it may have provided more guidance for algorithmic curation as Congress would have witnessed the rapid success of Google and its search prioritization algorithm, PageRank.)

To date, courts have mostly protected social media platforms under §230 even when they amplify unlawful and harmful content. Courts like the Second and Ninth circuits have been reluctant to draw a distinction between a website arranging content for a better user experience and a traditional publisher's editorial judgements.[198,199] They insist that holding platforms liable for recommendations generated by algorithms would effectively hold them liable for the underlying third-party (user-generated) content itself, which is precisely what §230 intended to prevent. On the other hand, a judicial minority (Katzmann, the Third Circuit) and lawmakers have argued that the recommendation algorithms go beyond passive facilitation and justify new legal accountability. They point to tangible harms that occur when algorithms prioritize the most extreme or dangerous material, ostensibly to maximize engagement. In their view, §230 should not be an all-encompassing shield in such contexts, especially when platforms may even profit from increased engagement with harmful content.

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[198] *Force supra* note 156.
[199] *Dyroff v. Ultimate Software Grp.*, 934 F.3d 1093 (9th Cir. 2019).

The defenses of §230 are robust. They caution that weakening immunity could significantly, if not completely impair platforms' abilities to offer users useful information. They argue that the vast scale of the Internet necessitates algorithms to sort information.[200] In today's context they maintain that algorithmic curation is protected under the First Amendment because it involves selecting, organizing, and prioritizing content to create a personalized user experience.[201] Their arguments emphasize that any incursion into §230 protection threatens their ability to moderate their platforms, thus impacting the quality and safety of information available to users.

From a bird's eye view, the jurisprudence around social media platforms is a result of user and lawmaker frustration with a perceived lack of accountability from social media companies. Legislatively, the U.S. is at a crossroads. Bills like EARN IT and SAFE TECH reflect a growing sentiment that some kind of recalibration is needed, whether to protect children, prevent harassment, or generally reign in platforms from treating the Internet like the Wild West.[202] However, finding space for regulation is not straightforward, given broad interpretations of §230, free speech protections, and the technical complexity and proprietary nature of the systems.

On the other hand, The European Union's Digital Services Act (DSA) mandates and enforces transparency through a risk-based approach, though its enactors did not have to deal with §230. The DSA does not dictate what content can and cannot be published; instead, it forces platforms to create processes and make content moderation transparent to users. It also requires them to seriously consider user safety concerns and the protection of fundamental rights.[203] Overall, the DSA attempts to keep the concept of basic immunity and thus prevent chilling, while holding them to a high standard of care so that algorithms are not black boxes optimizing solely for profit at society's expense. Realistically, current interpretations of §230 and America's robust free-speech protections make it unlikely for such sweeping regulation to take effect in the United States at present.

In the meantime, court decisions will continue to shape readings of §230. The Supreme Court has thus far avoided a direct ruling on algorithmic recommendations, but there is increasing

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[200] Ben Sperry, *Section 230 & Gonzalez: Algorithmic Recommendations Are Immune*, Truth on the Market (Feb. 1, 2023), https://truthonthemarket.com/2023/02/01/section-230-gonzalez-algorithmic-recommendations-are-immune/.
[201] *NetChoice, LLC v. Paxton* and *Moody v. NetChoice, LLC*, 603 U.S. 707 (2024).
[202] *Legislation to Reform Section 230 Reintroduced in the Senate, House*, Sen. Mark Warner (2023), https://www.warner.senate.gov/public/index.cfm/2023/2/legislation-to-reform-section-230-reintroduced-in-the-senate-house.
[203] Harsh V. Pachisia, *The EU's Digital Services Act Takes on "The Algorithm"*, Chi. Pol'y Rev. (Jan. 2024), https://chicagopolicyreview.org/2024/01/01/the-eus-digital-services-act-takes-on-the-algorithm/.

pressure for clarity as lower courts diverge.[204] A narrower Supreme Court interpretation of §230 would spur congressional action by "mobilizing the political influence of giant platforms." Though such a shift may risk destabilizing the Internet, destabilization may be necessary to create a more transparent social media landscape and information ecosystem online.[205]

[204] Cong. Rsch. Serv., *The United States Courts of Appeals: Background and Circuit Splits from 2024* (n.d.).
[205] Alan Z. Rozenshtein, *Interpreting the Ambiguities of Section 230*, Yale J. on Regul. (Apr. 2024).

Chapter 6: Remedies

Recommender systems fundamentally shape user experience on social media through algorithmic curation. Though these systems are extremely complex and rely on machine learning models with varied explainability, the overall themes of the output are definitively shaped by the business goals of the platforms. A "supply-and-demand" hypothesis offers an explanation for extremism on platforms: The amplification of "problematic" content reflects users' desire to consume such content, and the implementation of social media allows this demand to be met.[206]

The current legal status of algorithmic recommender systems remains unresolved. The Supreme Court has yet to rule on whether classifying modern algorithmic recommender systems as First Amendment-protected conduct make them ineligible for §230 protection, and in doing so suggest a path towards accountability. This tension stems from the hybrid nature of recommender systems in that they do reflect both editorial discretion (in deciding how content is selected and ranked) and automated, "neutral" conduct (in how algorithmic rules and systems govern those decisions). Platforms' opacity about the current implementation of their recommender systems further complicates attempts at legal classification, and analogies to historical platforms are weakened as systems become more complex and far-reaching. Buzzy terms like "machine learning" and "artificial intelligence" both overgeneralize and over specify, making it difficult for the general public to understand the extent to which their information consumption is shaped by individuals, not computers.

The outcome of this debate will influence not only platform accountability but also the future of content moderation, user autonomy, and online discourse. If SCOTUS rules that treating recommendation algorithms as First Amendment-protected speech renders them ineligible for §230 protection, lawmakers are still extremely limited in their ability to regulate the speech. However, they would be liable to civil suits such as in *Anderson v. TikTok*. If algorithms are not classified as First Amendment-protected speech, platforms will continue to receive wide-ranging §230 protection. Neither option seems to entirely solve the problem of platform accountability, which shapes how information circulates, how political and cultural narratives are formed, and how much power platforms wield over public discourse.

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[206] Kevin Munger & Joseph Phillips, *Right-Wing YouTube: A Supply and Demand Perspective*, 27 Int'l J. of Press/Pol. 1 (Oct. 2020), https://doi.org/https://doi.org/10.1177/1940161220964767.

Algorithmic recommender systems occupy a legally ambiguous space. They actively curate content and shape user feeds, but they also function as part of the overall infrastructure of their platforms. Since *Zeran,*[207] platforms have enjoyed wide ranging legal immunity under §230—even as they curate and amplify unlawful content. If recommender systems continue to receive §230 protection, they will rarely, if ever, be held accountable for algorithmically amplified harms such as the spread of misinformation, political extremism, and harassment. This broad immunity raises questions about platform accountability and responsibility. A particular one might be: How true to the original intentions of §230 are current interpretations by the Courts? Even *Zeran,* which established §230 as a "a great inducer of unfettered free speech," was a "bold and activist" interpretation of the meaning of the new legislation.[208] As recommender systems have become more sophisticated and involved in shaping user experiences, the passive conduit framework underpinning many interpretations of §230 has become increasingly difficult to justify in such general terms.[209]

On the other hand, abolishing §230 immunity entirely is likely to encourage platforms to engage in rigorous content vetting to avoid legal exposure, presenting a danger to free speech.[210] A possible remedy presents itself in a reinterpretation or amendment to the legislation that accounts for the personalized and more involved nature of modern recommender systems. Danielle Citron argues that §230(c)(1) immunity should no longer extend to platforms that "deliberately encourage, solicit, or maintain" unlawful content such as intimate privacy violations, cyberstalking, or harassment.[211] However, given the importance of Good Samaritan moderation, Citron proposes carving out bad actors and imposing a particularized *duty of care* for cases involving harmful abuse rather than eliminating immunity outright. A duty of care would require platforms to demonstrate that they took reasonable steps to mitigate foreseeable harm, such as employing trust and safety measures informed by industry best practices.[212] Such reforms would

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[207] *Zeran v. Am. Online, Inc.*, 129 F.3d 327 (4th Cir. 1997).
[208] Daniel Solove, *Restoring the CDA Section 230 to What It Actually Says*, TeachPrivacy (Feb. 4, 2021), https://teachprivacy.com/the-limits-of-section-230-accountability-for-algorithmic-decisions/.
[209] Daniel Solove, *The Limits of the CDA Section 230: Accountability for Algorithmic Decisions*, TeachPrivacy (August 29, 2024), https://teachprivacy.com/restoring-the-cda-section-230-to-what-it-actually-says/.
[210] Michael Daly Hawkins & Matthew J. Stanford, *Uproot or Upgrade? Revisiting Section 230 Immunity in the Digital Age*, U. Chi. L. Rev. (n.d.), https://lawreview.uchicago.edu/online-archive/uproot-or-upgrade-revisiting-section-230-immunity-digital-age.
[211] Danielle Keats Citron, *How to Fix Section 230*, 103 B.U. L. Rev. 713 (2023).
[212] *Id.*

leave space for free expression while ensuring that platforms are not free to design algorithms that amplify abuse without consequence.

The Digital Services Act (DSA) is an example of a regulatory framework that addresses this balance between expression and abuse by imposing specific obligations on online platforms regarding their recommender systems. Article 27 of the DSA mandates that platforms using recommender systems must "set out in their terms and conditions, in plain and intelligible language, the main parameters used in their recommender systems, as well as any options for the recipients of the service to modify or influence those main parameters."[213] Specifically, with "the existence of automated decision-making, including profiling [..] meaningful information about the logic involved, as well as the significance and the envisaged consequences of such processing for the data subject."[214] In other words, the platform is obligated to meaningfully explain how recommendations are made.

At this point, one might reasonably ask: What happens when these systems become too complex to understand? As shown in Chapter 4, there are a wide range of automated decision-making processes throughout recommender systems. Though driven by human action, the output of powerful, opaque models such as deep learning can be difficult to explain. It may appear that the balance between interpretability and predictive performance is a strict trade-off, but in practice this relationship is much more nuanced. Rather than interpretability and accuracy being inherently at odds, recent research suggests that factors such as user confusion, the design of the explanation interface, scalability, actionability, and even the framing of the task can all influence perceived explainability.[215, 216]

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[213] Regulation (EU) 2022/2065 of the European Parliament and of the Council of 19 October 2022 on a Single Market for Digital Services and Amending Directive 2000/31/EC (Digital Services Act), art. 27.
[214] *Id*. art. 13.
[215] Andrew Bell et al., *It's Just Not That Simple: An Empirical Study of the Accuracy-Explainability Trade-off in Machine Learning for Public Policy*, in *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT '22),* Association for Computing Machinery (2022), https://doi.org/10.1145/3531146.3533090.
[216] Moncef Garouani et al., *Investigating the Duality of Interpretability and Explainability in Machine Learning*, in *2024 IEEE 36th International Conference on Tools with Artificial Intelligence* (2024), https://doi.org/10.1109/ICTAI62512.2024.00125.

Bibal et al. distinguish between European transparency regulation for the private and public sectors, arguing that the former has much weaker requirements than the latter.[217] They present four levels of explainability for weaker, private sector legal requirements: key variables, all input variables used in a specific decision, how features are combined, and the full, interpretable model. In the public sector, entities are required to justify their decisions via jurisprudence and sometimes actual responses to user or legal arguments. It is straightforward to see how these kinds of transparency requirements could empower users to understand and potentially even control how content is recommended. Furthermore, the DSA obligates platforms to assess and mitigate risks associated with their recommender systems, such as potential impacts on mental health and the spread of harmful content. In the case of social media, European regulation aims to balance the benefits of personalized content delivery with protecting users from algorithmically amplified harms, serving as a model for amendments to §230.

Another approach to evolving recommender systems is giving users more direct control over how their feed is ranked. Currently, users have little agency over their feeds. Most platforms determine ranking criteria internally, limiting users' ability to modify how their content is organized. However, even when platforms allow some degree of customization, hyper-personalization may conflict with informed user consent because it relies on observed behavior rather than explicit feedback. Some platforms have already begun experimenting with offering users more direct control over feed makeup, but they are far from enough.[218,219]

Increased user controls over feeds and transparency represent a promising path toward improving engagement-based recommender systems, but there are two obstacles to the approach. On one hand, most users lack the technical knowledge to adjust algorithmic weights effectively and even if platforms introduce intuitive settings, users may still not understand how their choices affect their feeds. To address these challenges, education is an important part of transparency. On the other hand, platforms' financial incentives to maximize engagement for profit conflicts with allowing users to significantly reduce engagement-based ranking and could have a negative effect

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[217] Adrien Bibal et al., *Legal requirements on explainability in machine learning,* 29 Artif. Intell. Law (2021), https://doi.org/10.1007/s10506-020-09270-4.

[218] Adam Mosseri, *Control Your Instagram Feed with Favorites and Following*, Instagram (2022), https://about.instagram.com/blog/announcements/favorites-and-following.

[219] Jean Burgess et al., *Why Am I Seeing This Ad? The Affordances and Limits of Automated User-Level Explanation in Meta's Advertising System*, 26 New Media & Soc'y 2032 (Aug. 2024), https://doi.org/10.1177/14614448241251796.

on advertising revenue. Ideally, regulation could push platforms to develop algorithms and systems which optimize for engagement *and* are less harmful than current implementations.

Any alteration or introduction of regulation must pass judicial review. In the context of social media, if courts accept that recommender systems involve both expressive and non-expressive elements—design choices that structure societal discourse while also shaping technical behavior—regulations aimed at algorithmic curation and amplification of harms may qualify for intermediate scrutiny. Such laws would need to target the design or operation of the system, not the content of individual posts, and would have to be narrowly tailored to further substantial interests like protecting democratic integrity or individual privacy, without unnecessarily burdening expressive rights. It would be therefore prudent for regulatory efforts to include technical subject-matter experts for each component of the recommender system to ensure that any proposed measures are effective, appropriately tailored, and feasible.

Currently, there are few if any reasons platforms need to prioritize anything other than engagement on their platforms. As revenue-driven companies, profit serves as their primary measure of success. Modern social media recommender systems are designed to maximize engagement because that goal aligns directly with profit. Platforms like TikTok, Meta, and YouTube (Google) generate most of their revenue from advertising, and high engagement rates translate into higher ad impressions and more valuable ad placements. Despite popular concerns about addiction, child safety, polarization, misinformation, etc., platforms continue to enjoy massive user bases with minimal functional oversight. If §230 was amended to account for certain specific implementation details or results, platforms could face increased accountability for algorithmically amplified harms. Leaving the door open, even narrowly, to civil suits or regulatory action could create an incentive for platforms to improve their algorithms to reduce harm and improve user well-being.

Realistically, the financial risk of lawsuits would factor into a cost-benefit analysis for the companies. If risk is too great, platforms may resort to forgoing moderation or recommendation at all to regain Section 230 protection—leaving users to sort through massive amounts of irrelevant, unlawful content. This would bring us back to the pre-§230 days of "damned if you do moderate, damned if you don't." Therefore, platforms must be incentivized not to abandon content curation completely but to rethink their curation strategies. Ideally, platforms would develop recommender

systems that generate revenue *and* do not disproportionately amplify emotionally reactive, divisive and potentially dangerous content.

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Instead of engagement being the sole objective, platforms might use multi-objective optimization, where recommender systems are trained to balance engagement with other goals, such as content diversity, serendipity, and long-term user satisfaction. Already, popular pressure has pushed platforms to incorporate objectives beyond engagement, though to what extent and effect remains unclear.[220,221,222] Implementing multi-objective optimization raises technical challenges. As discussed in Chapter 3, obtaining engagement-based signals and metrics is computationally cheaper and more straightforward (since clicks and watch time are easy to measure), but optimizing for user satisfaction or content diversity requires more complex and subjective feedback mechanisms. Still, if there was a clear business incentive—in the form of financial risk from legal action, or compliance with clear legal parameters with the threat of monetary penalization—perhaps platforms could more meaningfully implement multi-objective optimization.

While platforms are unlikely to voluntarily adopt meaningful multi-objective optimization unless their financial or legal incentives shift, creating some sort of legal accountability might help shift those incentives. In such a case, platforms would face increased liability for algorithmic harms because their algorithms would not be protected under §230. Direct accountability on the part of the platforms likely would incentivize them to adopt an optimization strategy that serves business and social interests.

The value of human preference lies in its inexplicability. We are never entirely certain as to why we prefer something to another thing; if we could explain our preference precisely, why would anyone be curious about our preference and try it out themselves? As Dean Lomeli orates

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[220] Karen Hao, *YouTube Is Experimenting with Ways to Make Its Algorithm Even More Addictive*, MIT Tech. Rev. (Sept. 27, 2019), https://www.technologyreview.com/2019/09/27/132829/youtube-algorithm-gets-more-addictive/.
[221] Julia C. Wong, *Facebook Overhauls News Feed in Favor of 'Meaningful Social Interactions'*, The Guardian (Jan. 11, 2018), https://www.theguardian.com/technology/2018/jan/11/facebook-news-feed-algorithm-overhaul-mark-zuckerberg.
[222] Bringing the Magic of Friends Back to Facebook, *Meta* (Mar. 27, 2025), https://about.fb.com/news/2025/03/bringing-magic-of-friends-back-to-facebook/.

in the novel *Conclave*, "Certainty is the great enemy of unity. Certainty is the deadly enemy of tolerance."[223] Machines are certain in some ways, in the sense that their preferences are dictated by a set of predetermined technical processes. In other ways, machine learning is an inherently unstable process, as models are constantly updating their outputs based on new data. If you tell a large language model it is wrong, (as of today) it will not fight you. It will correct itself—because it is learning.

The legal status of AI-powered recommender systems is neither a simple question, nor a static one. Section 230 was enacted to address a newly developing Internet, and for almost three decades after its enactment its language was sufficiently broad to cover the rapid technological development which occurred. When technical systems were simpler, broad or imprecise language was often enough to convey their function. But with the rise of today's sophisticated personalized recommender systems, that language cannot meaningfully describe the realities of algorithmic curation. Going forward, we must employ holistic decision-making that considers technology, culture, and jurisprudence to determine the legal status of complex artificial intelligence-powered recommender systems. This is not an issue that can be dismissed or further put off as Americans increasingly depend on social media for virtually all facets of their daily lives.[224] If courts continue to classify recommender systems as broadly protected by Section 230, Congress may find it prudent to amend the legislation to better fit the modern systems of today. Judges may be moved to reinterpret the legislation to better apply to the modern systems of today. Therefore, legal and regulatory entities have a critical and temporal obligation to strengthen their understanding of modern recommender systems on social media.

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[223] Robert Harris, *Conclave* (2016).

[224] Pew Research Center, *Social Media*, https://www.pewresearch.org/topic/internet-technology/platforms-services/social-media/ (last visited Apr. 4, 2025).